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   - Introduction to CUDA, SGEMM, and the Naive Kernel
   - L1 Cache and Basic Tiling
   - Throughput, Arithmetic Intensity, and the Roofline Model
   - Applying the Roofline Model: Block Tiling and Thread Coarsening

Section Overview

1. Optimizing **Ge**neral **M**atrix **M**ultiplication on CUDA
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What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

$$C \leftarrow \alpha AB + \beta C$$

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- Other values let one call scale the product and accumulate it into an existing matrix.

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- Setting $\alpha = 1$ and $\beta = 0$ gives ordinary matrix multiplication: $C = AB$.
- Other values let one call scale the product and accumulate it into an existing matrix.

Example

During the forward pass when training a model, we accumulate the gradients from the entire batch.

What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

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- Setting $\alpha = 1$ and $\beta = 0$ gives ordinary matrix multiplication: $C = AB$.
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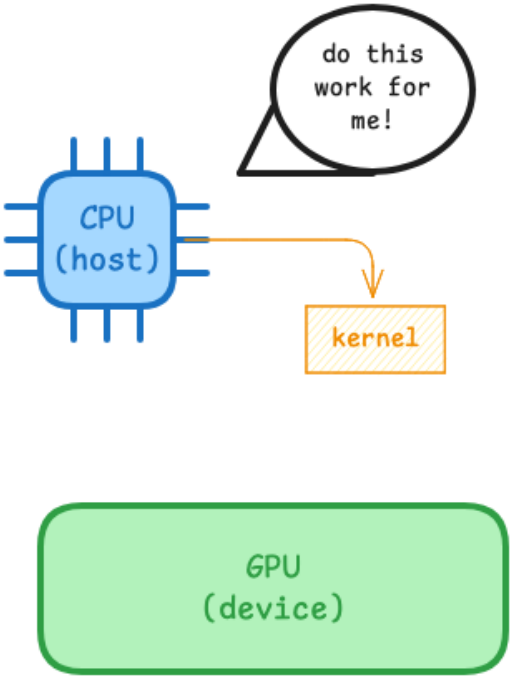
```
sgemm(
  int M, int N, int K,
  float alpha, float *A, float *B,
  float beta, float *C
)
```

$$\underbrace{A}_{M \times K} \quad \underbrace{B}_{K \times N} \quad \underbrace{C}_{M \times N}$$

- M: rows of A and C
- N: columns of B and C
- K: shared reduction dimension

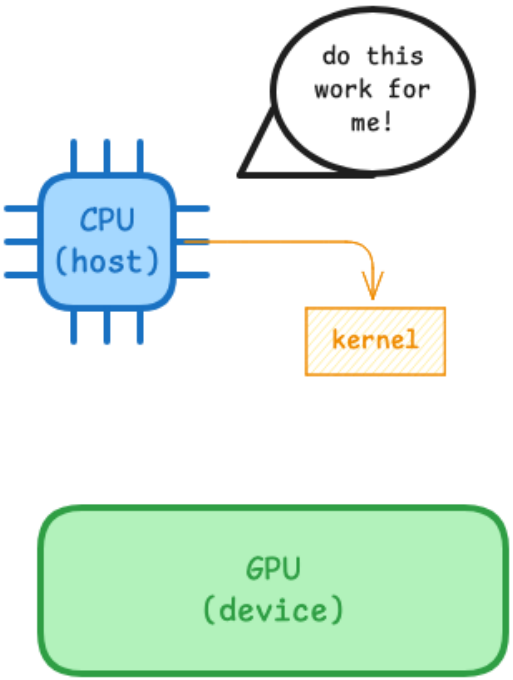
CUDA Basics (Host vs. Device)

- The CPU is known as the **host**, and manages top level logic.
- The GPU is known as the **device**, and does the heavy lifting.
- The host launches **kernels** to the device.



CUDA Basics (Host vs. Device)

- The CPU is known as the **host**, and manages top level logic.
- The GPU is known as the **device**, and does the heavy lifting.
- The host launches **kernels** to the device.




```
int main() {
  const int M = 128, N = 128, K = 128;

  float *dA, *dB, *dC;
  cudaMalloc(&dA, M * K * sizeof(float);
  cudaMalloc(&dB, K * N * sizeof(float));
  cudaMalloc(&dC, M * N * sizeof(float));

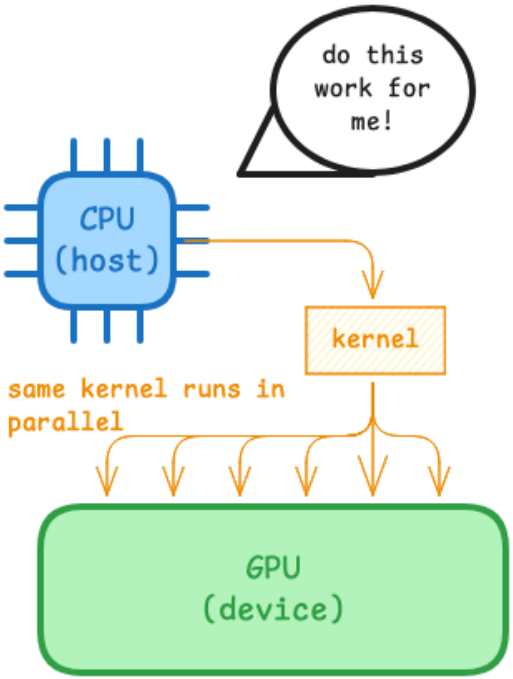
  // load values into these device pointers...

  run_sgemm_naive(M, N, K, 1, dA, dB, 0, dC);

  cudaFree(dA); cudaFree(dB); cudaFree(dC);
}
```

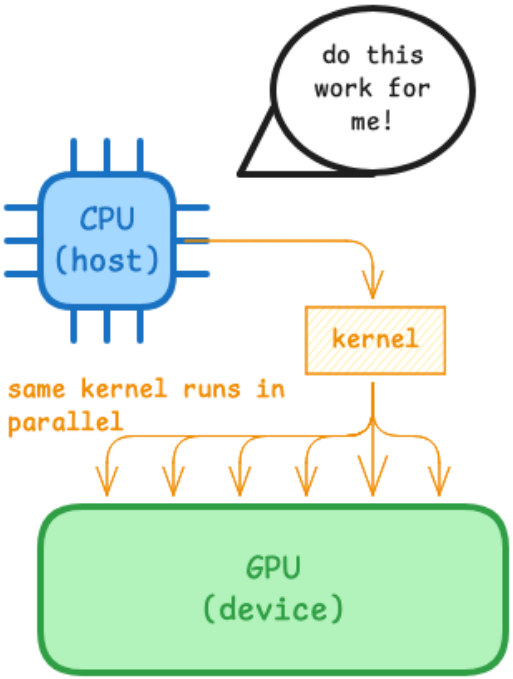
CUDA Basics (Kernel Execution)

- A kernel runs in parallel on many **threads** on the device.
- The number of threads the device allocates is configured by the host, when it **launches** the kernel.



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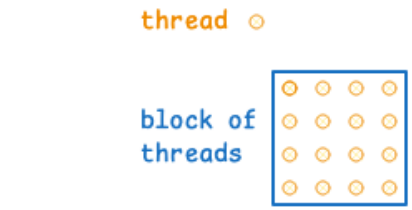


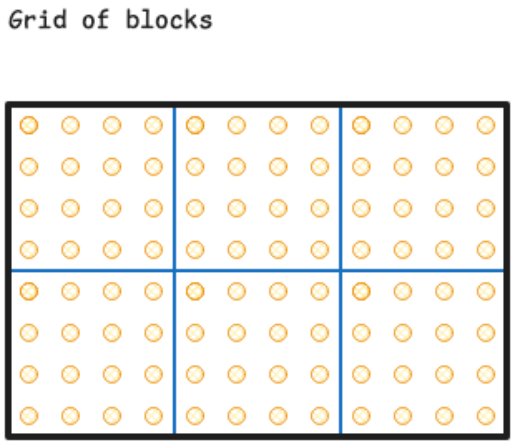

```cpp
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  // implementation hidden for now ...
}
```

```cpp
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
```

CUDA Basics (Threads, Blocks, Grids)

- Threads are grouped into **blocks**.
- The collection of blocks requested by the host is organized in a **grid**.






```cpp
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  // implementation hidden for now ...
}
```

```cpp
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
```

Why Threads, Blocks, Grids?

This seems complicated, why can't we just ask for N threads?

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Relation to hardware

This hierarchy is closely related to how the device schedules the execution. We will see later how to use this to optimize performance.

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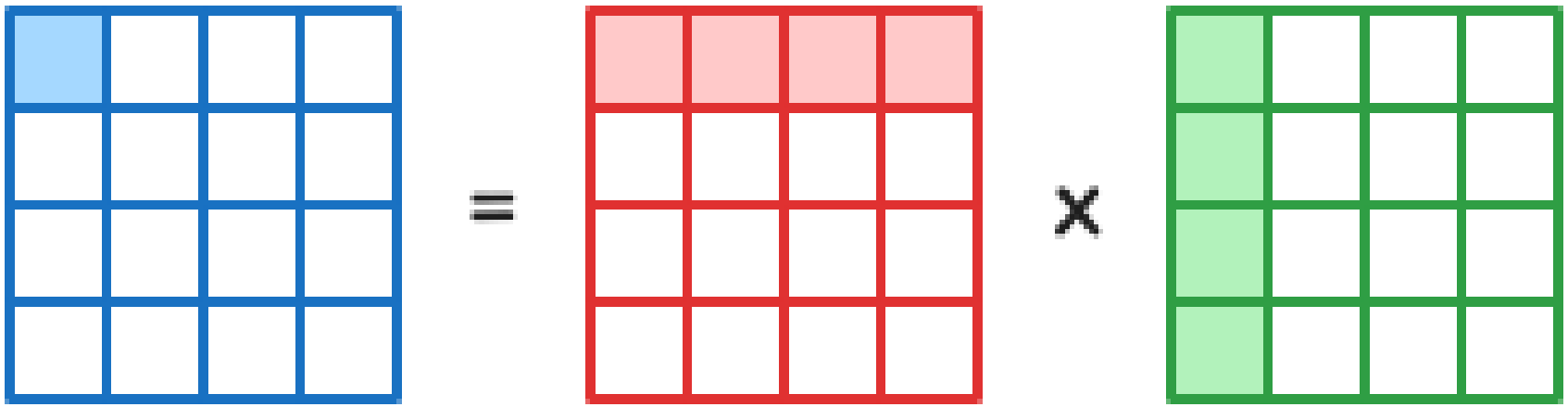
Thread, Block, Grid hierarchy

Threads are grouped into blocks of constant size/shape.
Blocks are launched as a grid.
We choose the dimensions of the grid to control how many blocks are launched, and thus how many threads are launched.

Let's design a SGEMM kernel!

Recall

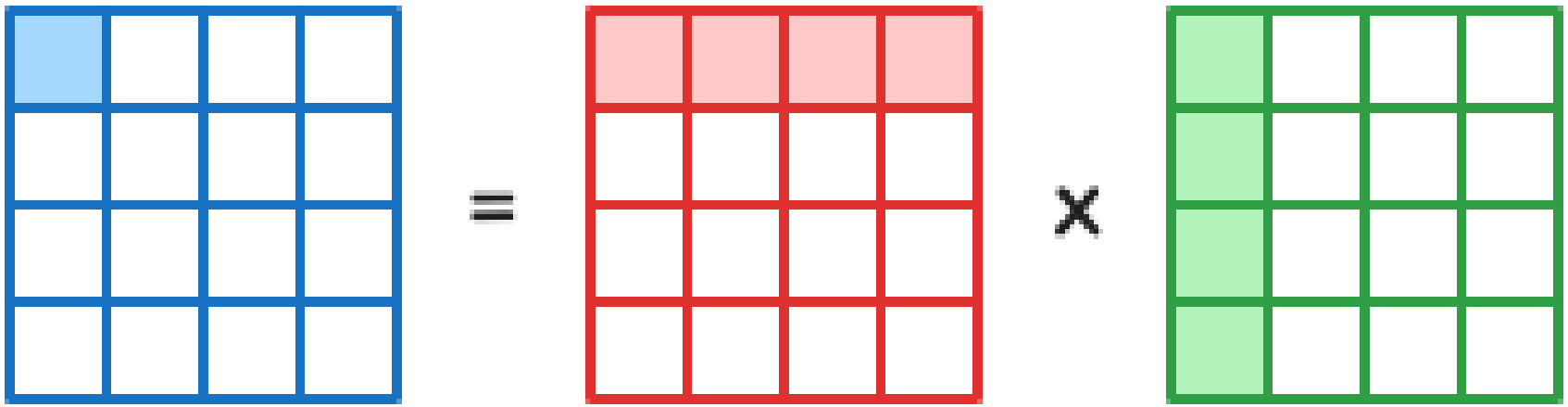
Recall that matrix multiplication is naturally parallel, and the output elements of C can be computed independently in parallel.



Let's design a SGEMM kernel!

Recall

Recall that matrix multiplication is naturally parallel, and the output elements of C can be computed independently in parallel.



We can naturally map one thread to each output element!

Naive kernel launch

Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C.

Naive kernel launch

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We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000.

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Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000. Thus, C must have shape 5000×4000.

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Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000. Thus, C must have shape 5000×4000. $m = 5000$, $n = 4000$, $k = 3000$.

Naive kernel launch

Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000. Thus, C must have shape 5000×4000. $m = 5000$, $n = 4000$, $k = 3000$.

We want blocks of 32×32, so $BM, BN = 32$.

```
int M = 5000, N = 4000, K = 3000;
float alpha = 1, beta = 0;

float *dA, *dB, *dC; // Allocate buffers on the DEVICE.
cudaMalloc(&dA, M * K * sizeof(float);
cudaMalloc(&dB, K * N * sizeof(float));
cudaMalloc(&dC, M * N * sizeof(float));
// BM, BN are compile-time constants!
launch_sgemm_naive<32,32>(M, N, K, alpha, dA, dB, beta, dC);
```

Naive kernel launch

Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000. Thus, C must have shape 5000×4000. $m = 5000$, $n = 4000$, $k = 3000$.

We want blocks of 32×32, so $BM, BN = 32$.

```
launch_sgemm_naive<32,32>(M, N, K, alpha, dA, dB, beta, dC);
// inside launcher function
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C)
```

Naive kernel launch

Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000. Thus, C must have shape 5000×4000. $m = 5000$, $n = 4000$, $k = 3000$.

We want blocks of 32×32, so $BM, BN = 32$.

```
launch_sgemm_naive<32,32>(M, N, K, alpha, dA, dB, beta, dC);
// inside launcher function
  dim3 blockDim(32, 32);
  dim3 gridDim((4000 + 32 - 1) / 32, (5000 + 32 - 1) / 32); // Ceil division
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C)
```

Naive kernel launch

Example (One thread per output element)

We want to do a plain matrix multiplication between matrices A and B and store it into C. Let A have shape 5000×3000, and B have shape 3000×4000. Thus, C must have shape 5000×4000. $m = 5000$, $n = 4000$, $k = 3000$.

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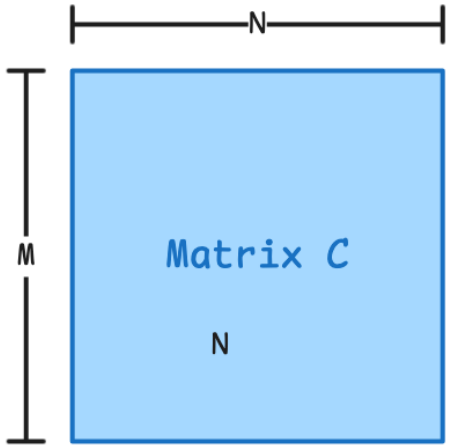
```
launch_sgemm_naive<32,32>(M, N, K, alpha, dA, dB, beta, dC);
// inside launcher function
  dim3 blockDim(32, 32);
  dim3 gridDim(126, 158); // Ceil division
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C)
```

Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

  if (y < M && x < N) {
    float sum = 0;
    for (int i = 0; i < K; i++) {
      sum += A[y * K + i] * B[i * N + x];
    }

    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
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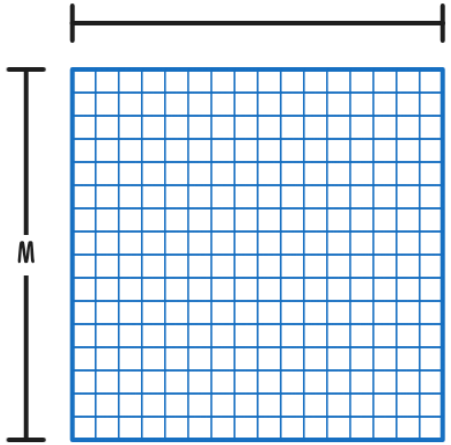


Naive SGEMM Kernel

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                            const float alpha, const float *A, const float *B,
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  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

  if (y < M && x < N) {
    float sum = 0;
    for (int i = 0; i < K; i++) {
      sum += A[y * K + i] * B[i * N + x];
    }

    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
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  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
```

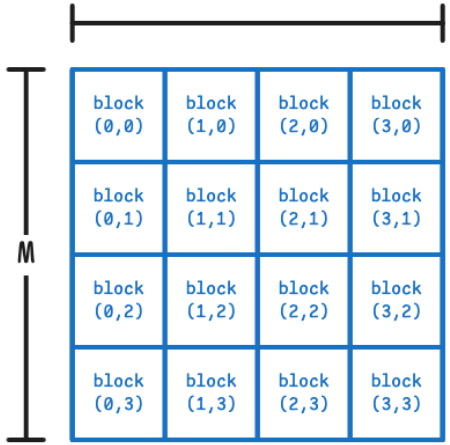


Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

  if (y < M && x < N) {
    float sum = 0;
    for (int i = 0; i < K; i++) {
      sum += A[y * K + i] * B[i * N + x];
    }

    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
```

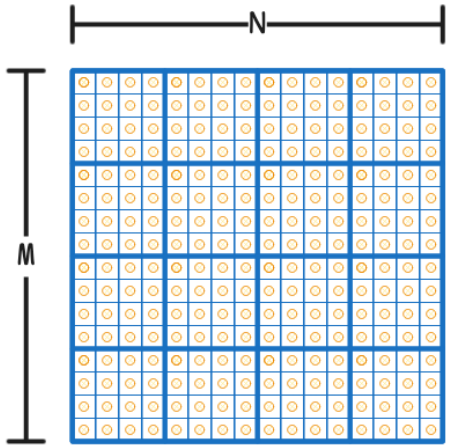


Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

  if (y < M && x < N) {
    float sum = 0;
    for (int i = 0; i < K; i++) {
      sum += A[y * K + i] * B[i * N + x];
    }

    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
```

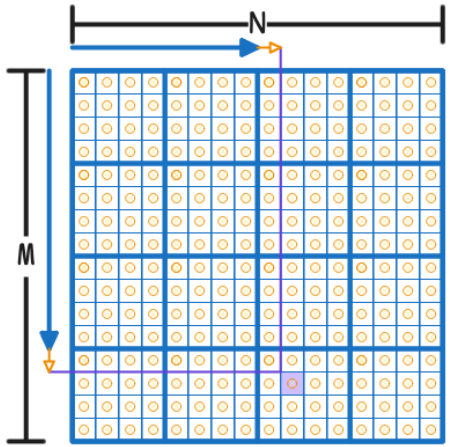


Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

  if (y < M && x < N) {
    float sum = 0;
    for (int i = 0; i < K; i++) {
      sum += A[y * K + i] * B[i * N + x];
    }

    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
template <int BM, int BN>
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  dim3 blockDim(BN, BM);
  dim3 gridDim((N + BN - 1) / BN, (M + BM - 1) / BM);
  sgemm_naive<<<gridDim, blockDim>>>(M, N, K, alpha, A, B, beta, C);
}
```

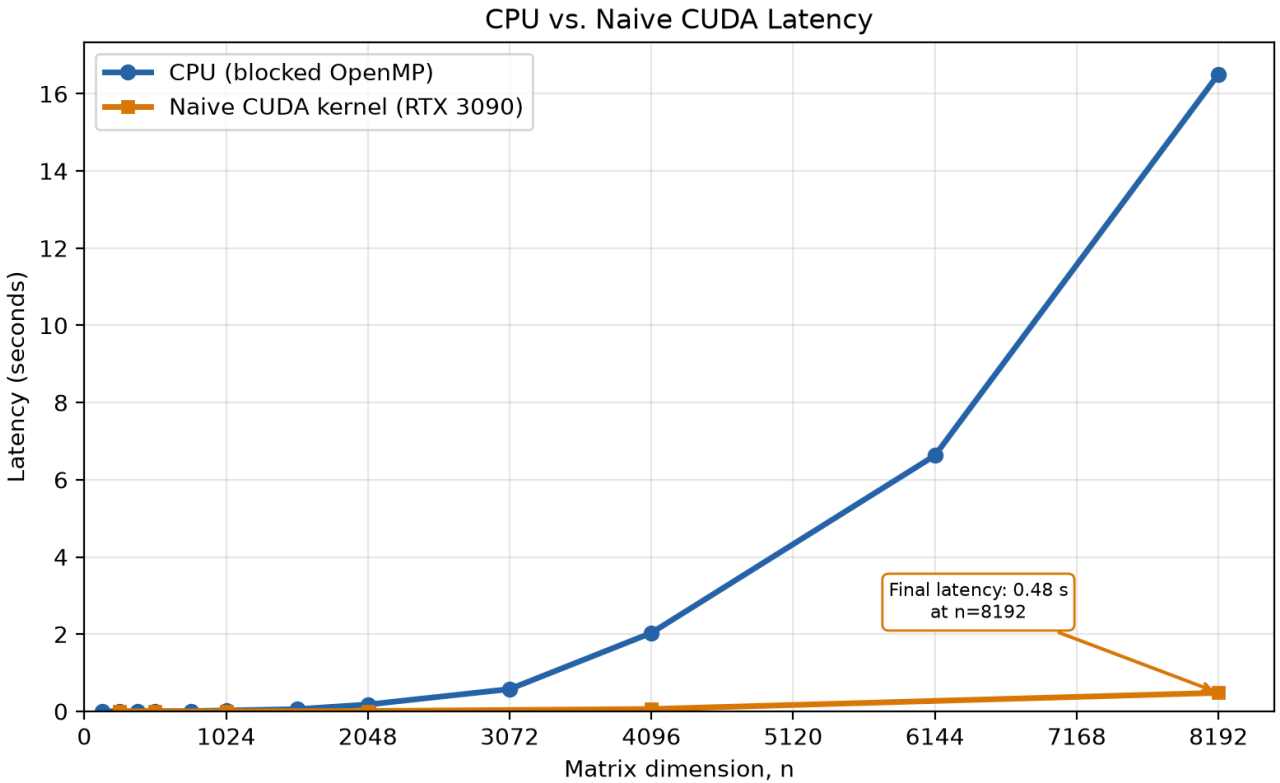


Naive SGEMM visualization

- One thread computes one element of C.
- It walks one row of A and one column of B for the full dot product.

Play naive-kernel visualization

Naive CUDA is already faster than the CPU

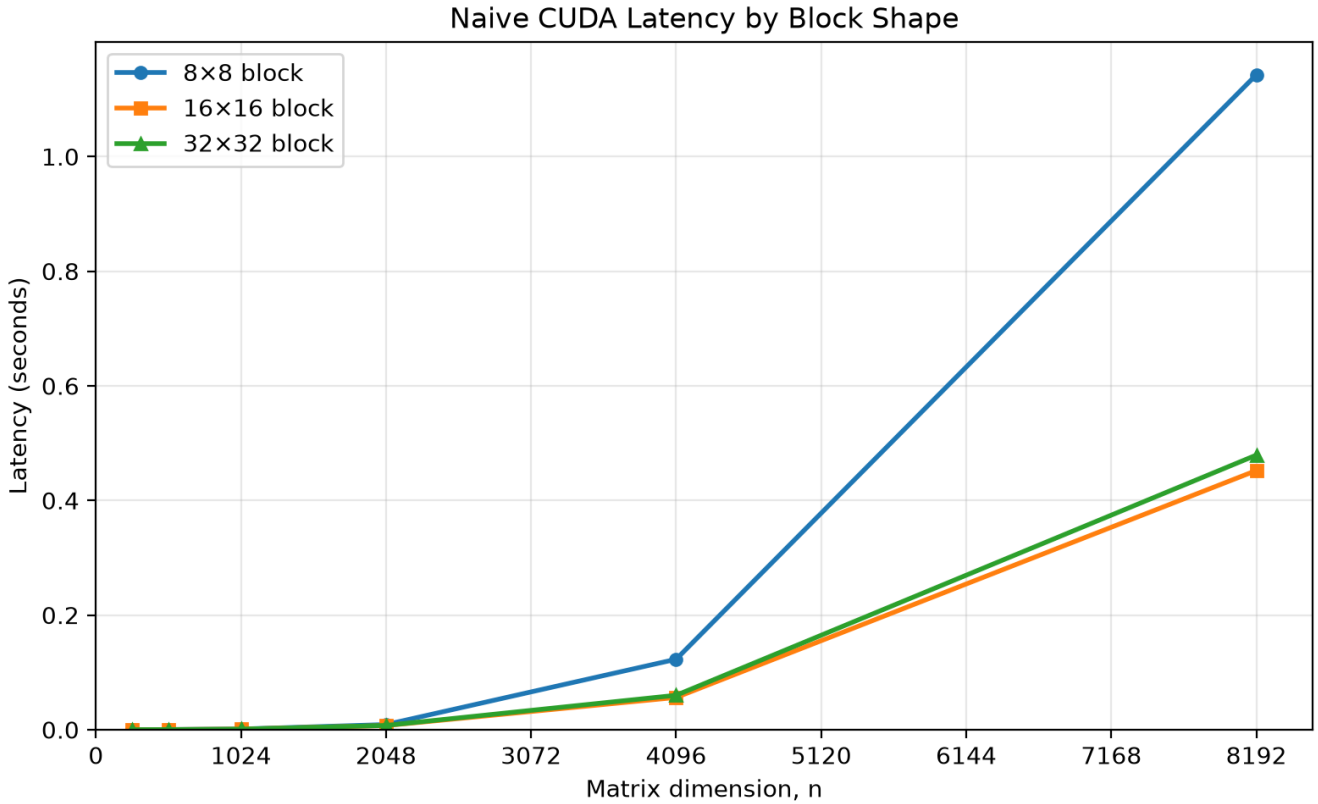


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Let's do some experimenting

This kernel assigns one thread per output element, and computes them independently. So changing block size shouldn't matter right?



Blocks and SMs

Why does block size matter?

This happens because entire **blocks** are **arbitrarily** scheduled to SM's.
The larger block sizes of 16x16 and 32x32 lead to more **L1 cache hits**.

Blocks and SMs

Why does block size matter?

This happens because entire **blocks** are **arbitrarily** scheduled to SM's.
The larger block sizes of 16x16 and 32x32 lead to more **L1 cache hits**.

Block shape	**L1 Cache Hit Rate**
8×8	83.51%
16×16	87.13%
32×32	94.98%

RTX 3090, $M = N = K = 4096$

Shared Memory (SMEM) Caching

Programmable L1 Cache (SMEM)

Remember how our launch config required us to group threads as blocks? That is because **blocks** are scheduled to SM's as a unit, so **threads in the same block have access to the SM's L1 cache/SMEM.**

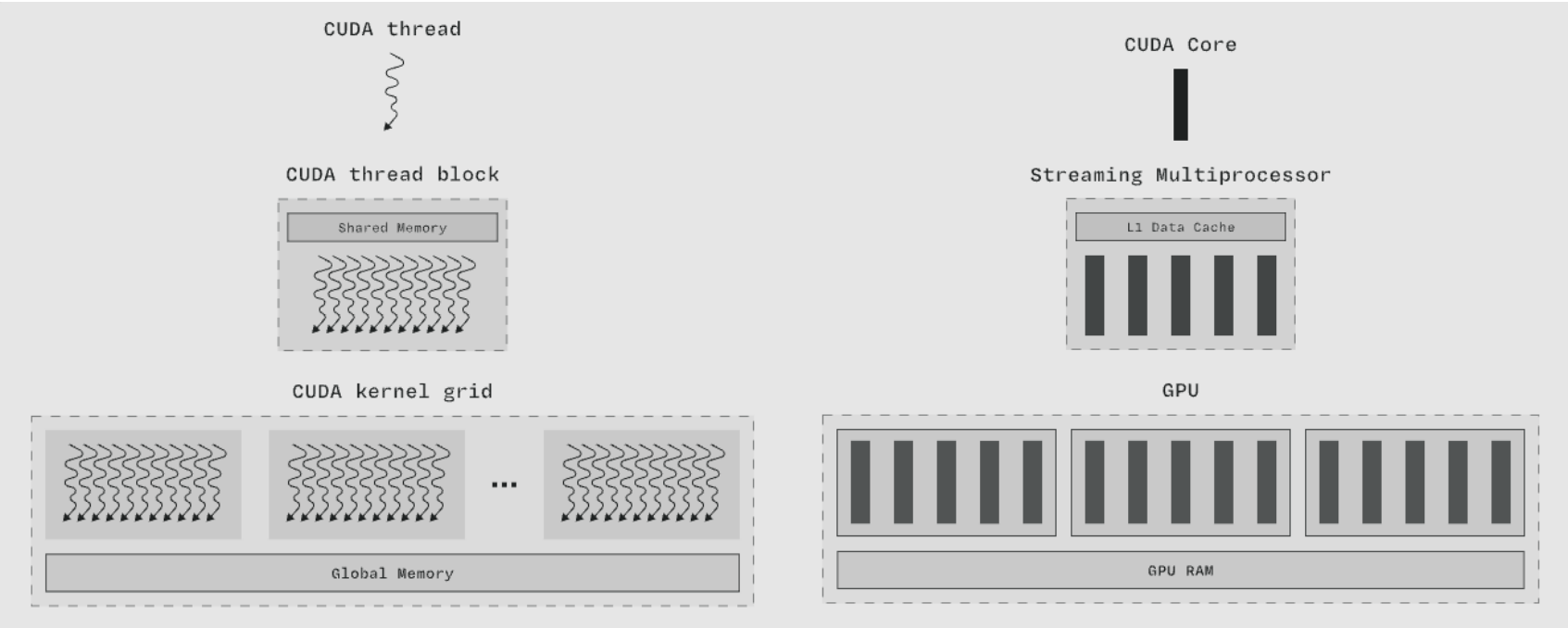


Image source: Modal, "CUDA Programming Model," *GPU Glossary*.

Tiling

Explicit caching of data shared between threads.

Main Idea

Before doing any computation, load a chunk of the necessary data from DRAM to SMEM as a block. Then, memory accesses for the computation now only go to SMEM.

Basic tiled SGEMM: step by step

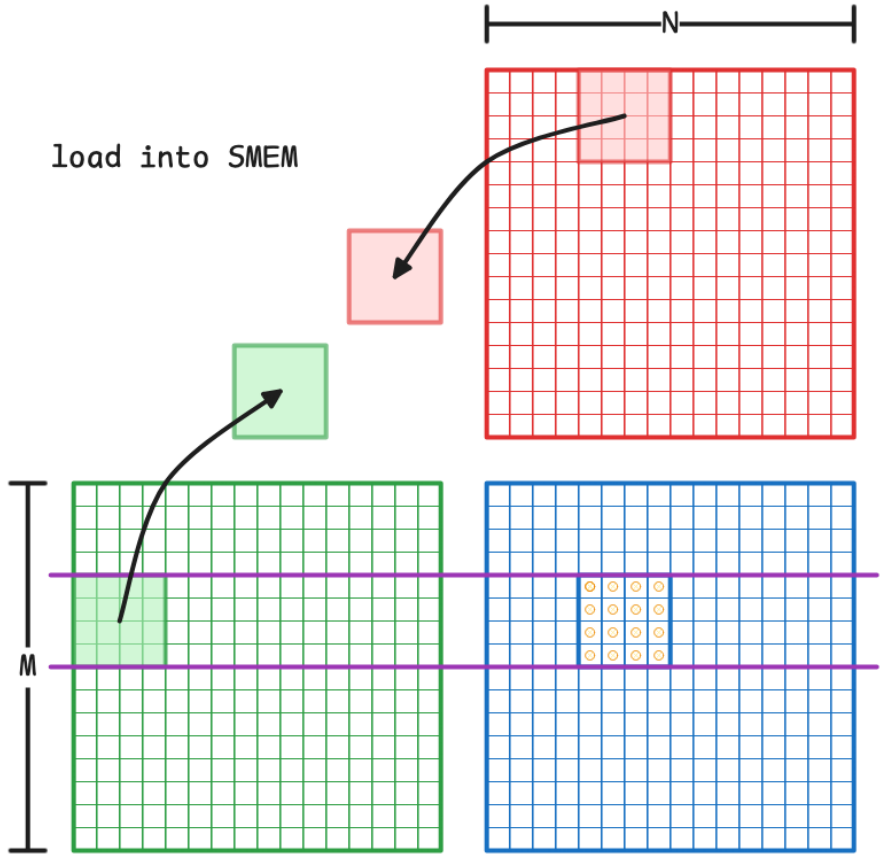
```
__shared__ float As[BK][BK];
__shared__ float Bs[BK][BK];
float sum = 0.0;

for (int k = 0; k < K; k += BK) {

  const int A_tile_x = k + xt;
  if (y < M && A_tile_x < K) {
    As[yt][xt] = A[y * K + A_tile_x];
  } else {
    As[yt][xt] = 0.0;
  }

  const int B_tile_y = k + yt;
  if (B_tile_y < K && x < N) {
    Bs[yt][xt] = B[B_tile_y * N + x];
  } else {
    Bs[yt][xt] = 0.0;
  }
  __syncthreads();

  for (int i = 0; i < BK; i++) {
    sum += As[yt][i] * Bs[i][xt];
  }
  __syncthreads();
}
```



Basic tiled SGEMM: step by step

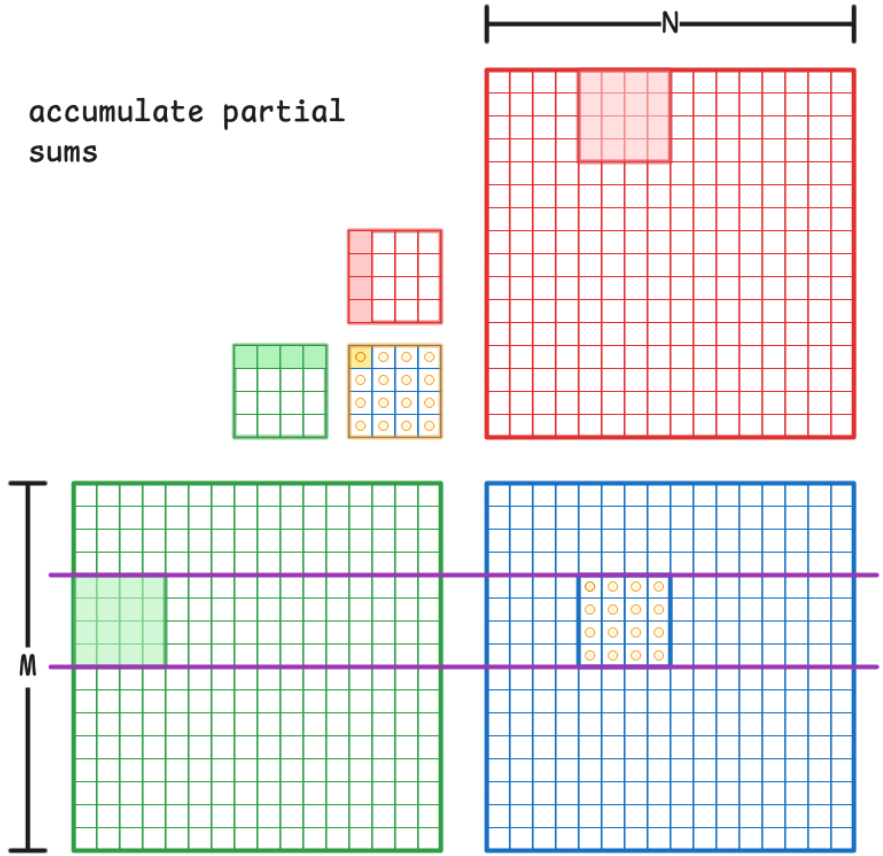
```
__shared__ float As[BK][BK];
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float sum = 0.0;

for (int k = 0; k < K; k += BK) {

  const int A_tile_x = k + xt;
  if (y < M && A_tile_x < K) {
    As[yt][xt] = A[y * K + A_tile_x];
  } else {
    As[yt][xt] = 0.0;
  }

  const int B_tile_y = k + yt;
  if (B_tile_y < K && x < N) {
    Bs[yt][xt] = B[B_tile_y * N + x];
  } else {
    Bs[yt][xt] = 0.0;
  }
  __syncthreads();

  for (int i = 0; i < BK; i++) {
    sum += As[yt][i] * Bs[i][xt];
  }
  __syncthreads();
}
```



Basic tiled SGEMM: step by step

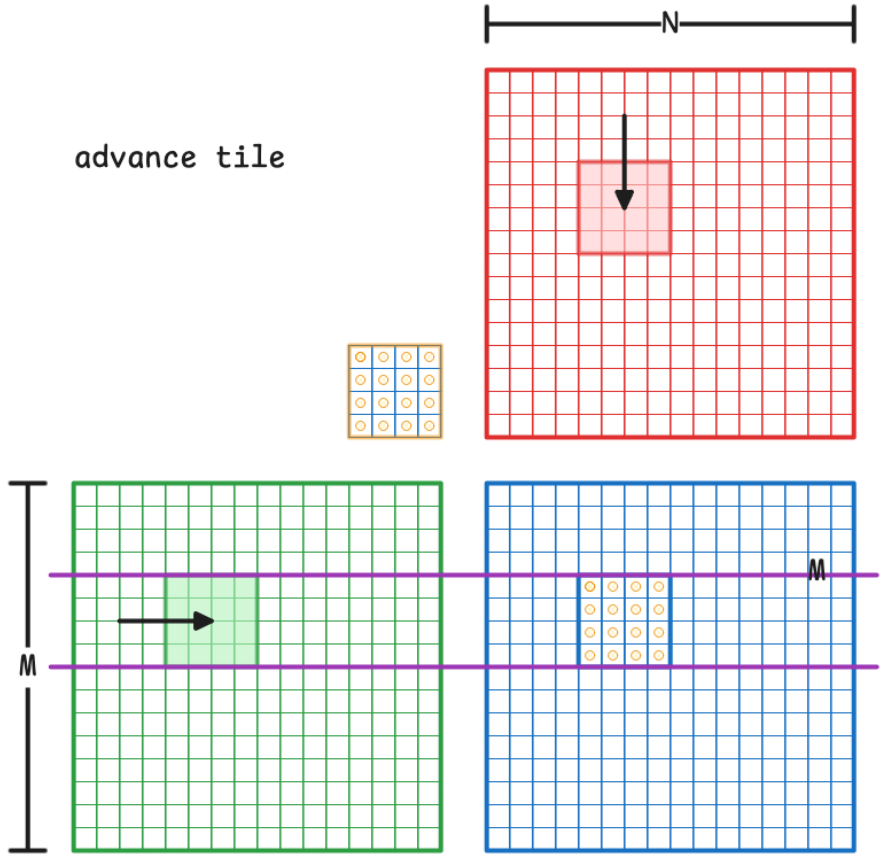
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  for (int i = 0; i < BK; i++) {
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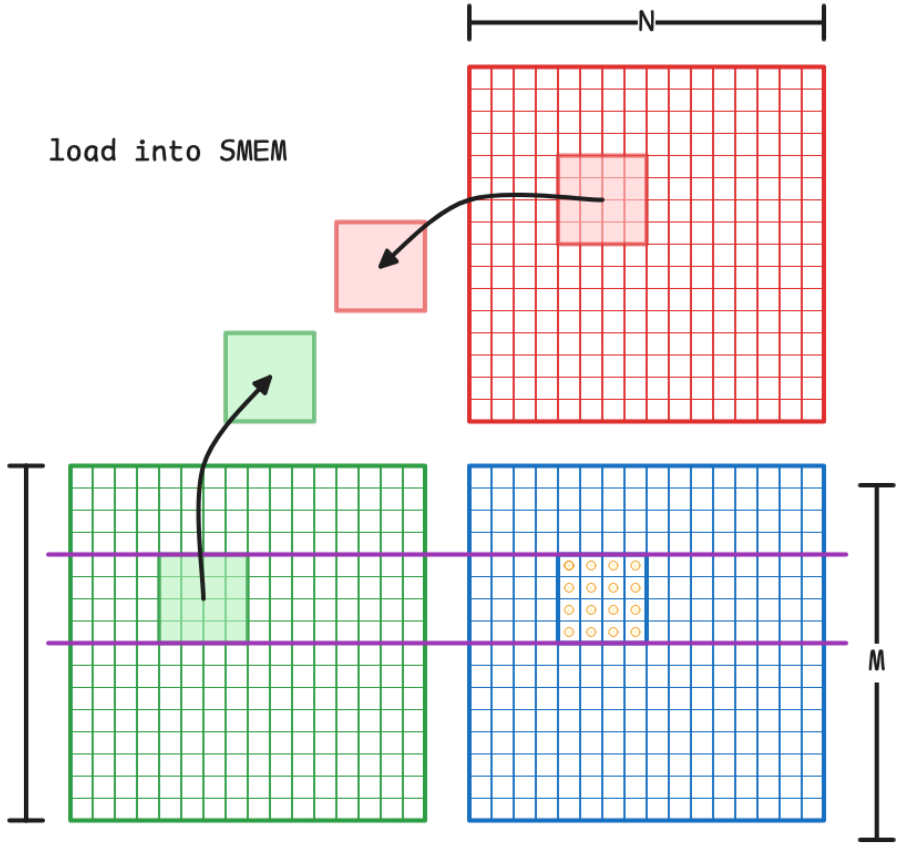
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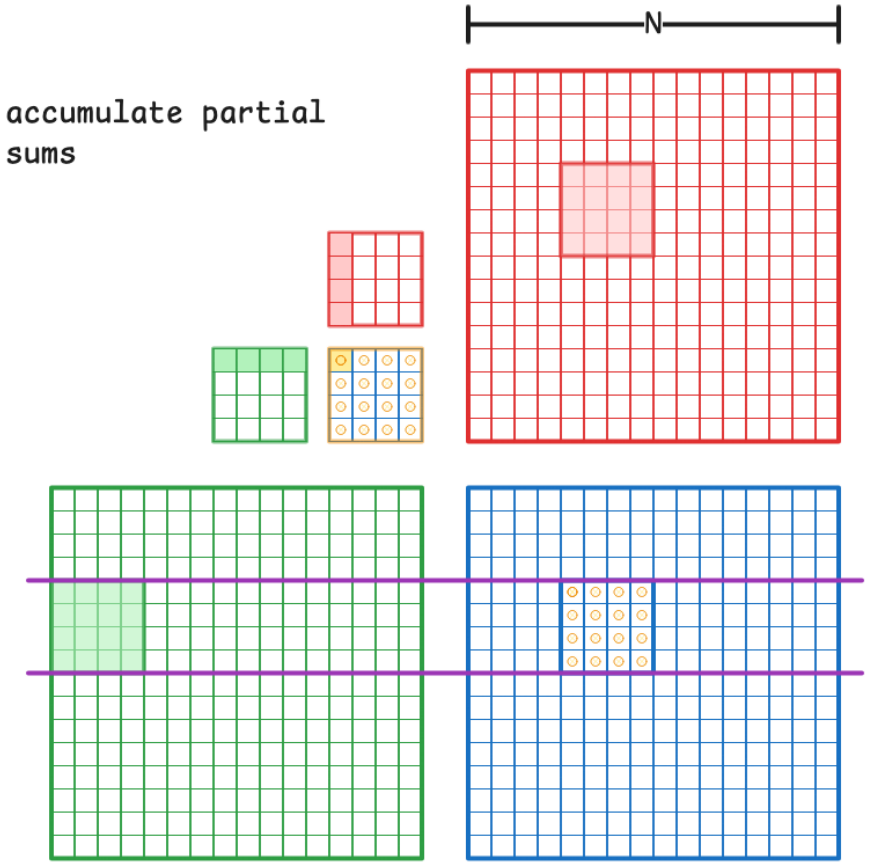
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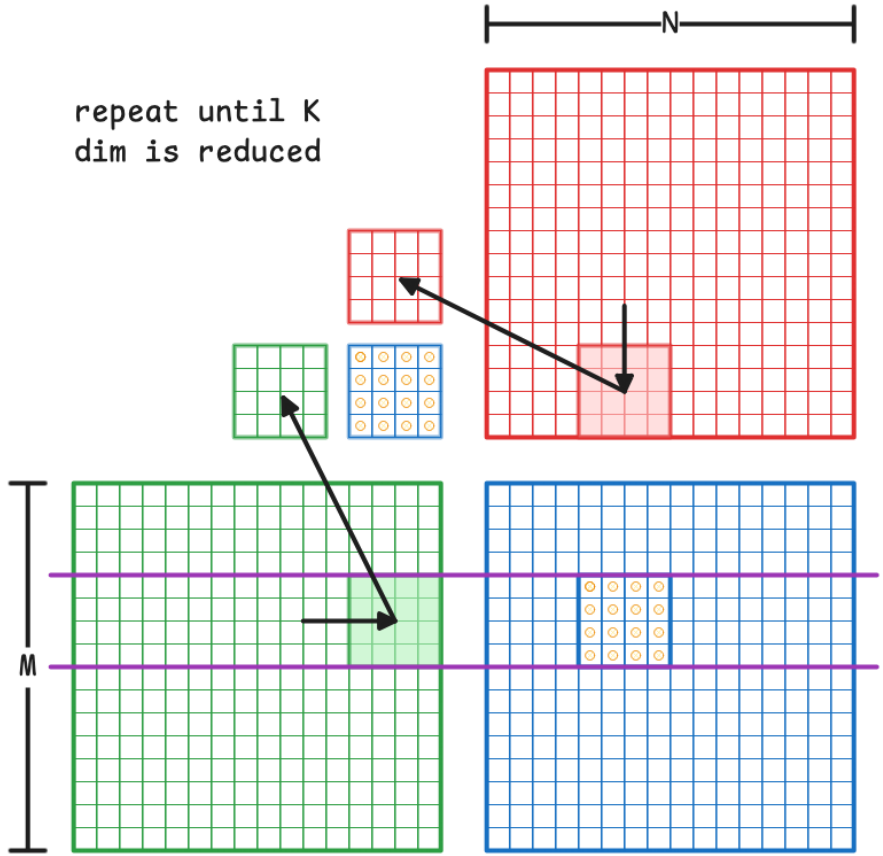
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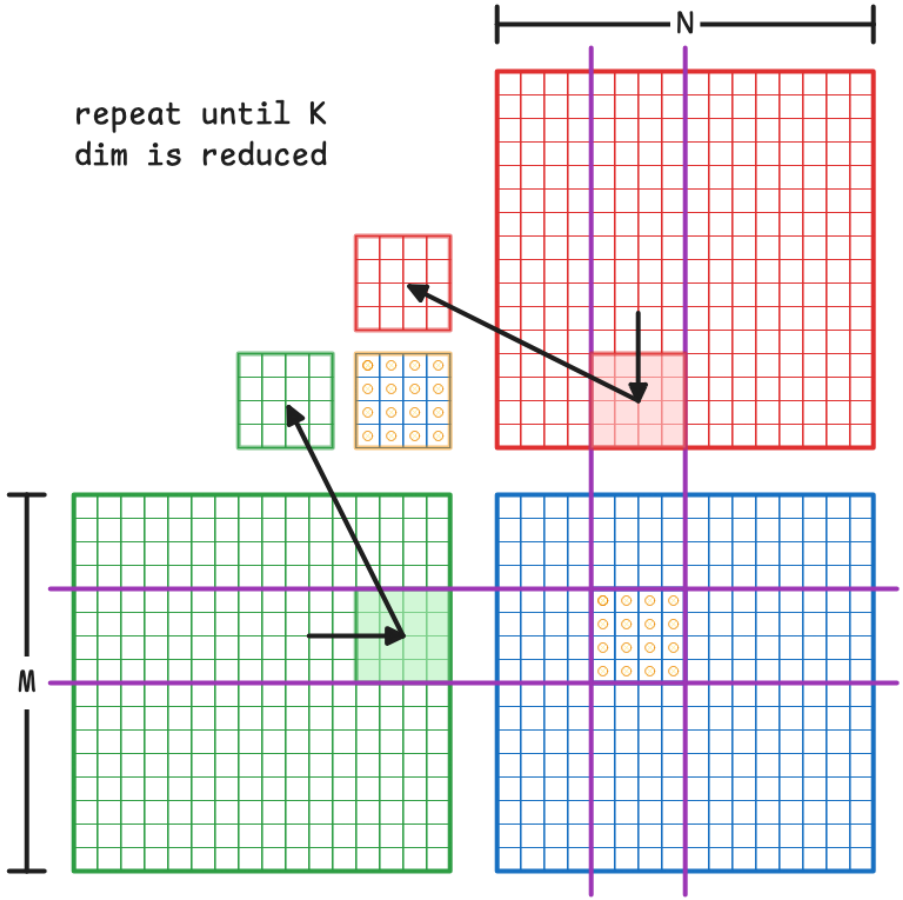
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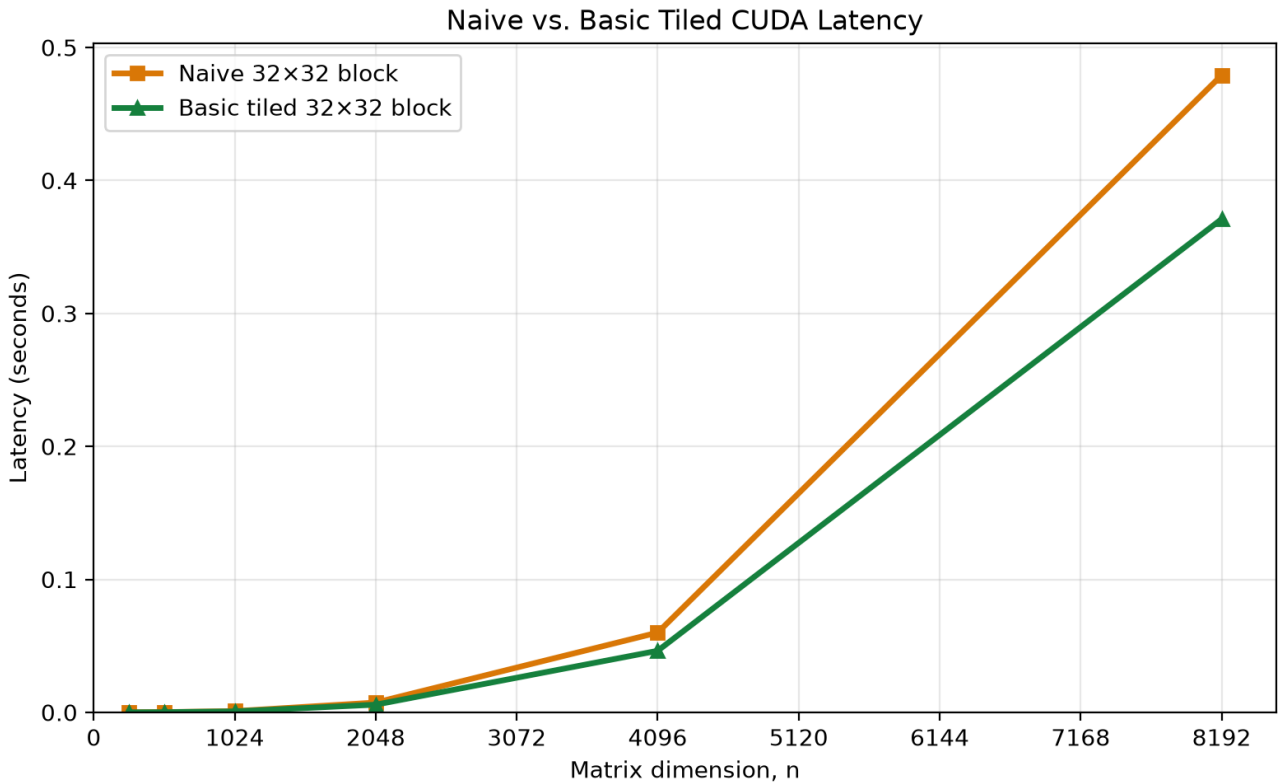


Basic tiled SGEMM visualization

- Watch each block load small regions of A and B together.
- The loaded tiles are reused to update an entire tile of C.
- Synchronization separates loading, computing, and advancing to the next tiles.

Play tiled-kernel visualization

Tiling performance



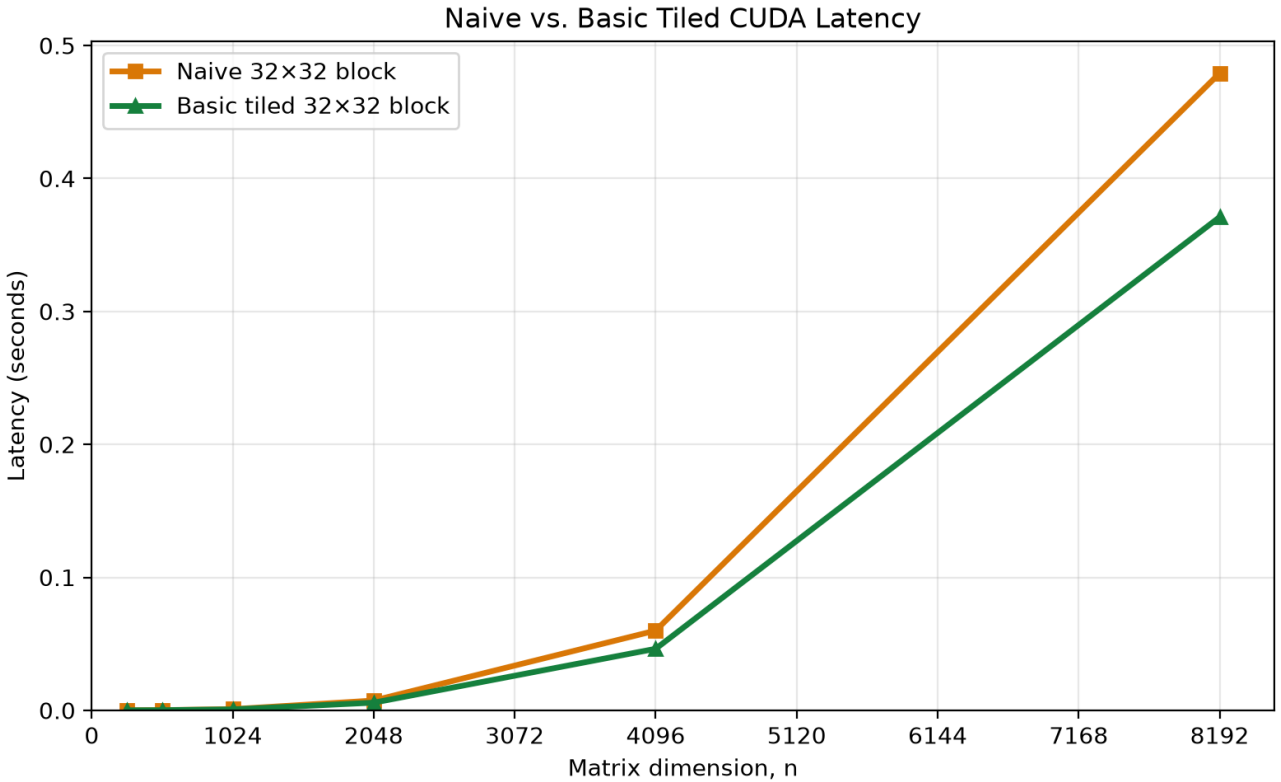
At $n, m, k = 8192$, tiled 32×32 latency is 371 ms, a 22.6% improvement over the naive.

Section Overview

1. Optimizing **Ge**neral **M**atrix **M**ultiplication on CUDA
   - Introduction to CUDA, SGEMM, and the Naive Kernel
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   - Throughput, Arithmetic Intensity, and the Roofline Model
   - Applying the Roofline Model: Block Tiling and Thread Coarsening

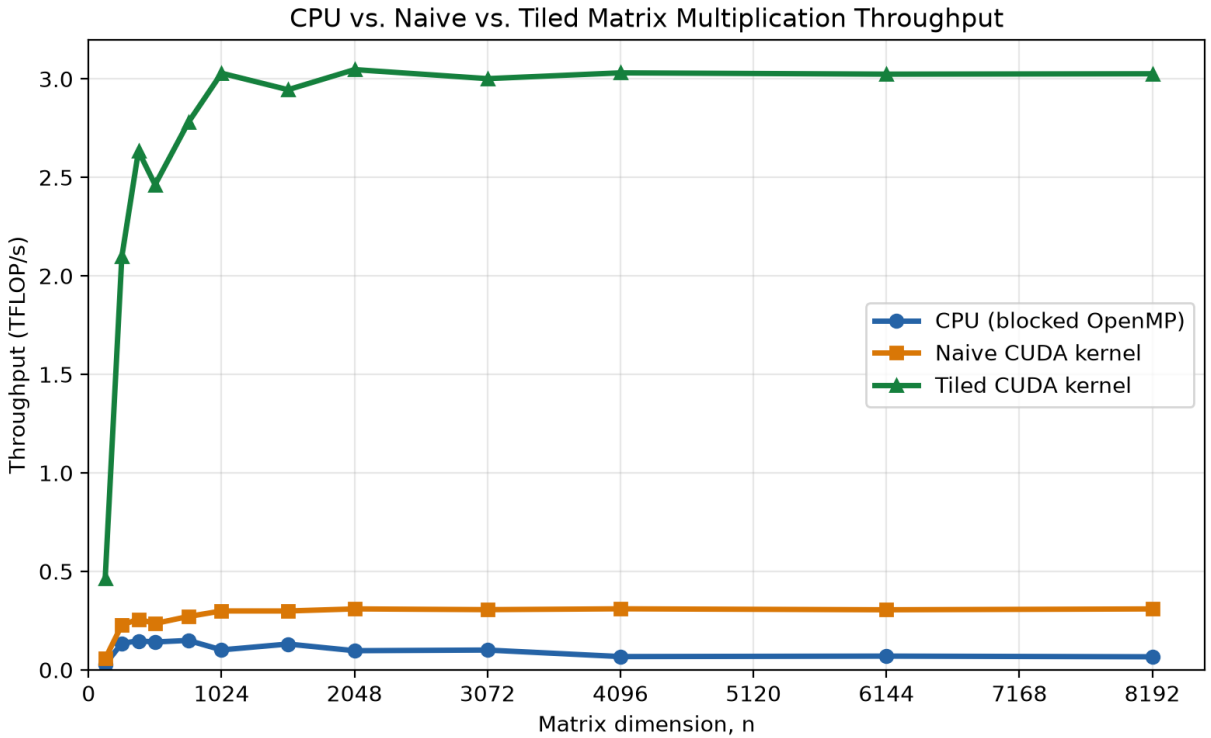
Why latency alone isn't enough

- Latency tells us how long one kernel launch takes, but it depends on the problem size.



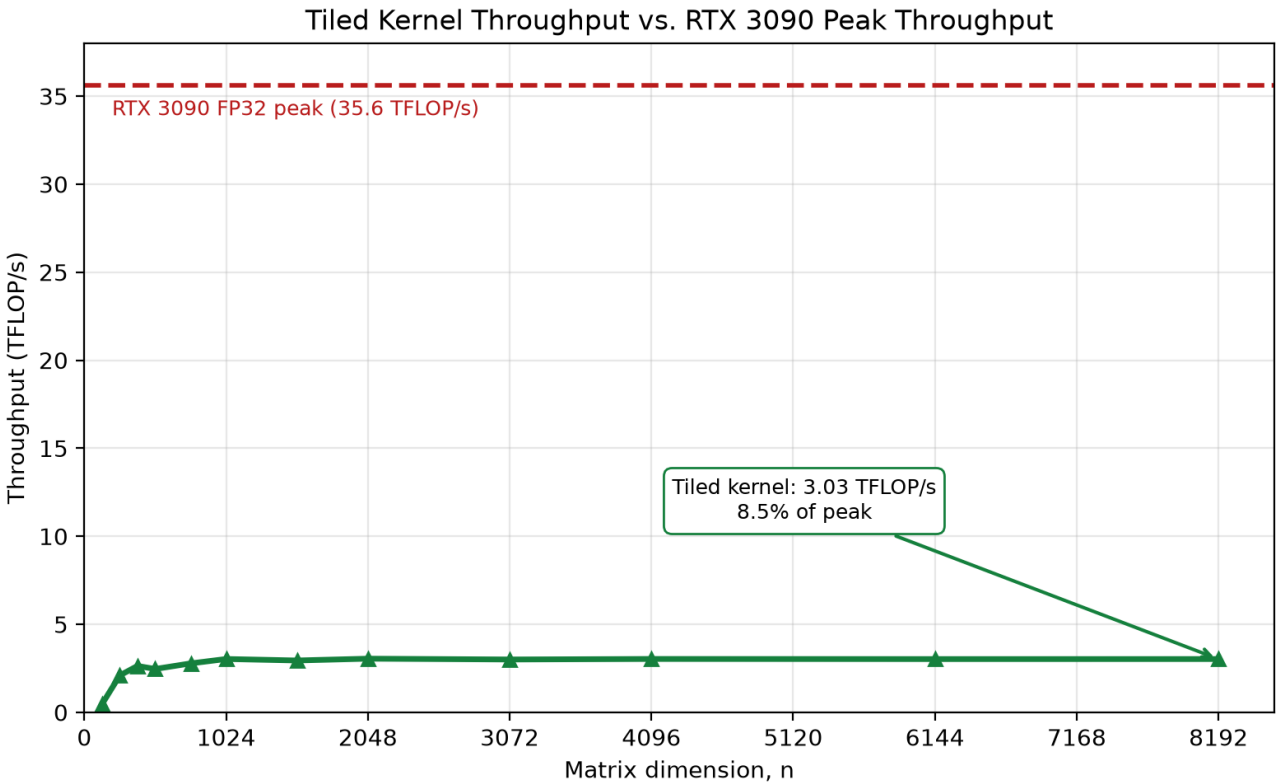
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- To compare performance **across problem sizes**, we need to account for both the work completed and the time required. This gives us **achieved throughput**.



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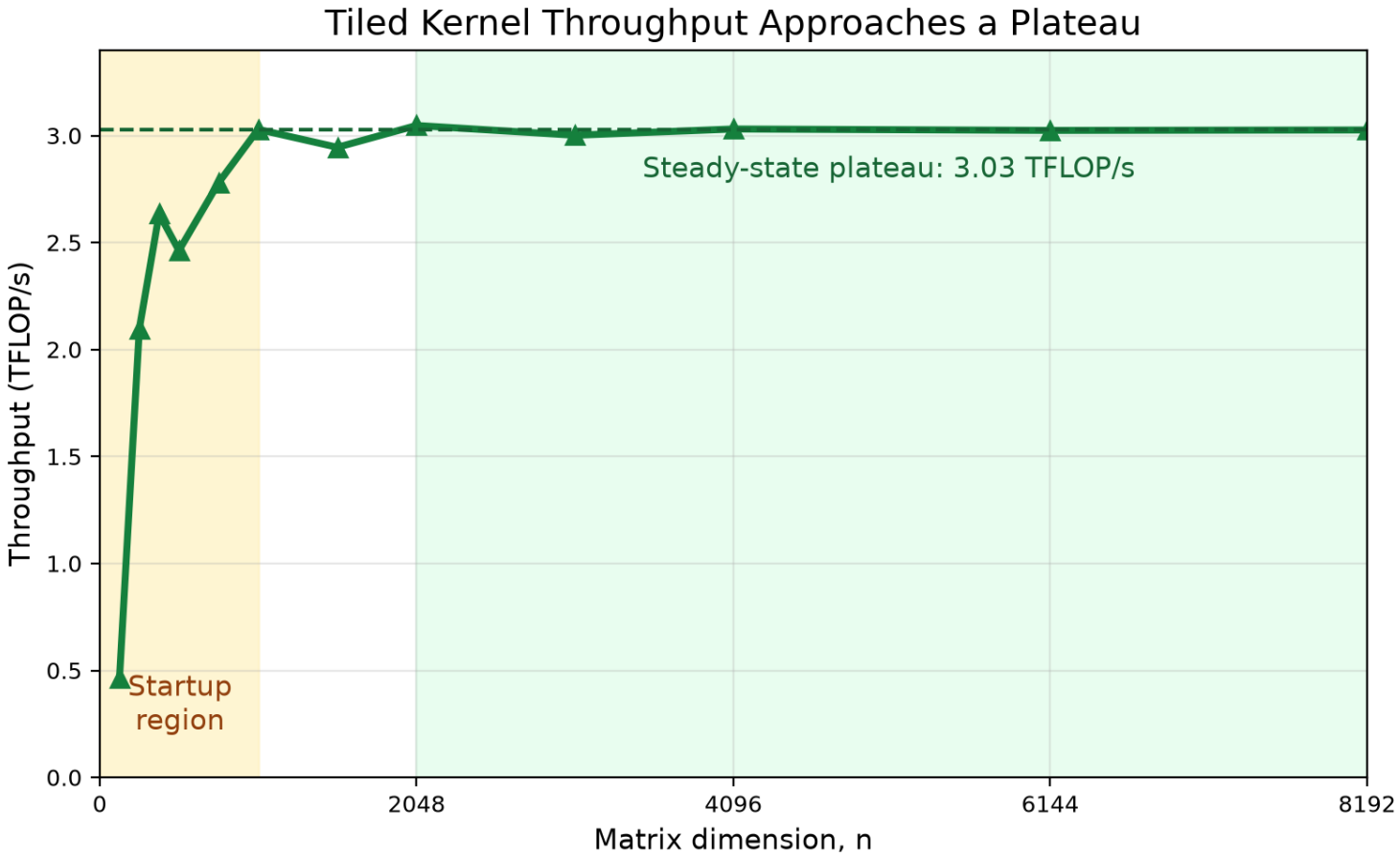
Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

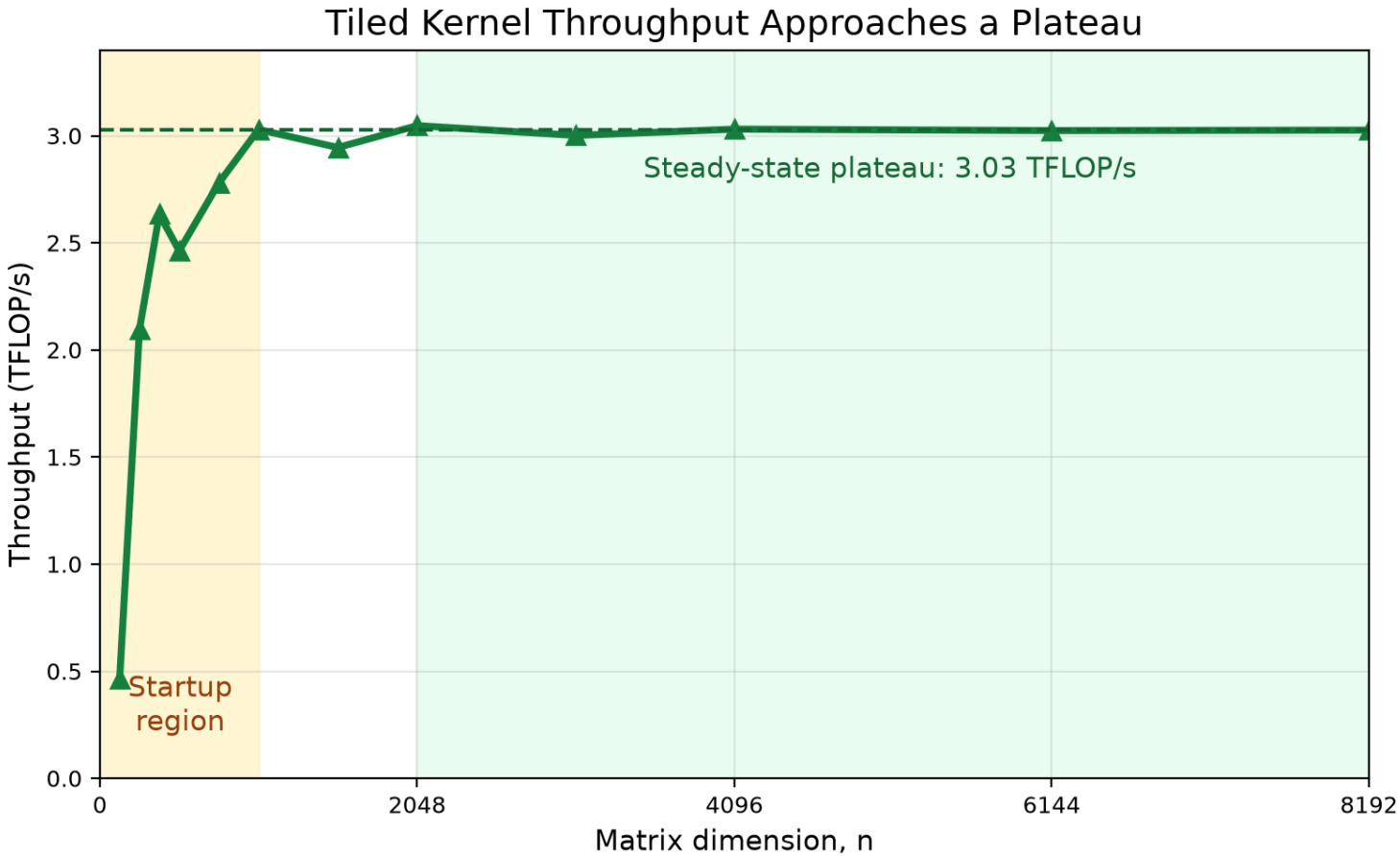
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Normalizing by useful work makes comparisons fairer, not size-independent.



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Measurement rule

Measure in the plateau region, and report the problem size with the result.

Throughput as a performance measure

Throughput

Throughput is the total useful work (FLOPs) done per unit time. *We previously discussed theoretical peak throughput of hardware.*

$$\text{Throughput} = \frac{\text{FLOPs}}{\text{time}}$$

Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput.

Arithmetic Intensity

Arithmetic Intensity

We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

$$\text{AI} = \frac{\text{FLOPs}}{\text{bytes moved}}$$

Example

For $n, k, m = 8192$, our tiled kernel moved a total of 137.7 GB of data from DRAM. With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \times 10^{12} \text{ FLOP}}{137.7 \times 10^{9} \text{ bytes}} \approx \boxed{8.0 \text{ FLOP/byte}}.$$

Two Key Constraints of Hardware

Our hardware is characterized by

1. Peak throughput: 35.6 TFLOP/s
2. DRAM bandwidth: 935.8 GB/s

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At 935.6 GB/s of bandwidth, our **memory roof** is

$$935.8 \text{ GB/s} \times 8.0 \text{ FLOP/byte} \approx 7.49 \text{ TFLOP/s}.$$

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This is much lower than the peak throughput of 35.6 TFLOP/s. **No matter how we optimize this kernel, without increasing AI, we will never achieve greater than 7.49 TFLOP/s**.

Hypothetical Kernel

Our hardware is characterized by

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Example

Imagine we have some kernel that has double the AI at 16.0 FLOP/s.

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$$935.8 \text{ GB/s} \times 16.0 \text{ FLOP/byte} \approx 15.0 \text{ TFLOP/s}.$$

Since $15.0 \text{ TFLOP/s} < 35.6 \text{ TFLOP/s}$, this kernel will still be memory bound.

Hypothetical Kernels

Our hardware is characterized by

1. Peak throughput: 35.6 TFLOP/s
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AI (FLOP/byte)	Upper bound throughput (TFLOP/s)
8	7.49
16	14.97
32	29.95
36	33.69
38	35.56
40	37.43

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40	37.43

Wait! Our peak throughput is 35.6 TFLOPs! At AI $\geq$ 38, we are **compute bound**. Our data can finally keep up with the CUDA Cores!

Ridge Point

Our hardware is characterized by

1. Peak throughput: 35.6 TFLOP/s
2. DRAM bandwidth: 935.8 GB/s

AI = 38 FLOP/byte is known as the **ridge point** *RP*, and can be found by solving:

$$\text{Data roof} = \text{Peak throughput}$$

$$\text{DRAM Bandwidth} \times \text{RP} = \text{Peak throughput}$$

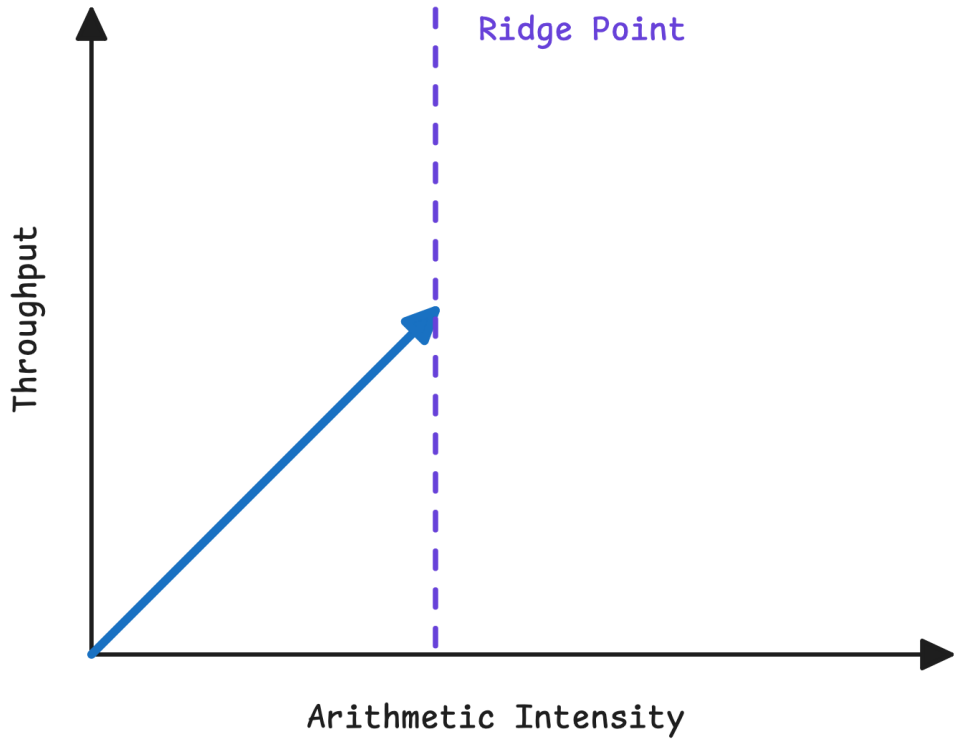
$$\text{RP} = \frac{\text{Peak throughput}}{\text{DRAM Bandwidth}}$$

$$\text{RP} = \frac{35.6 \text{ TFLOP/s}}{935.8 \text{ GB/s}} \approx 38.0 \text{ FLOP/s}$$

Roofline Model

Roofline Model

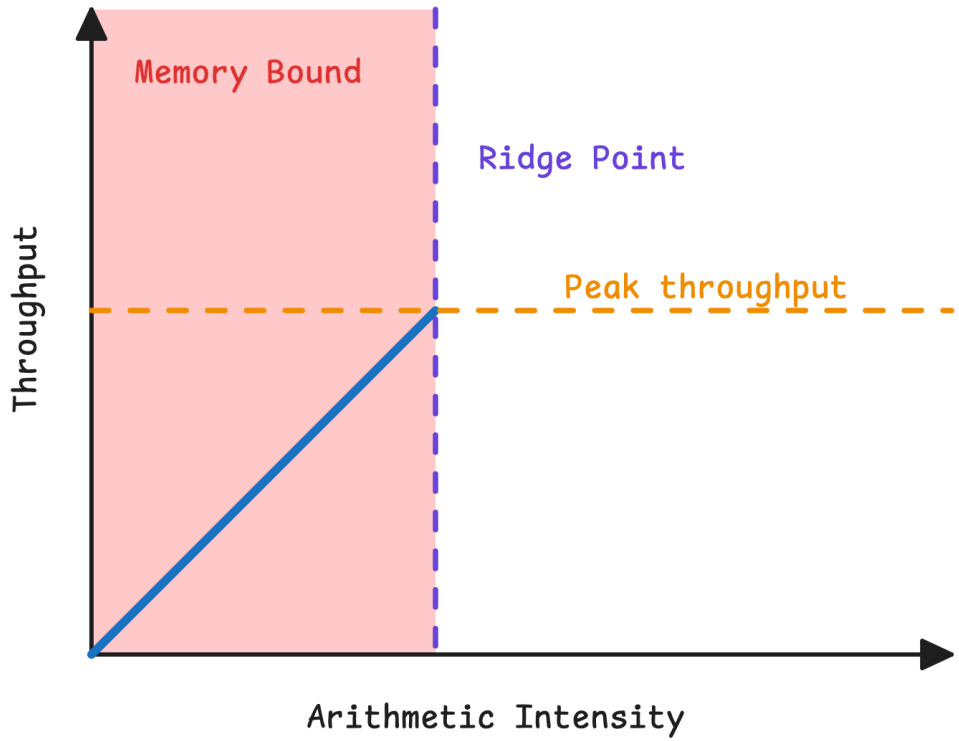
For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI.



Roofline Model

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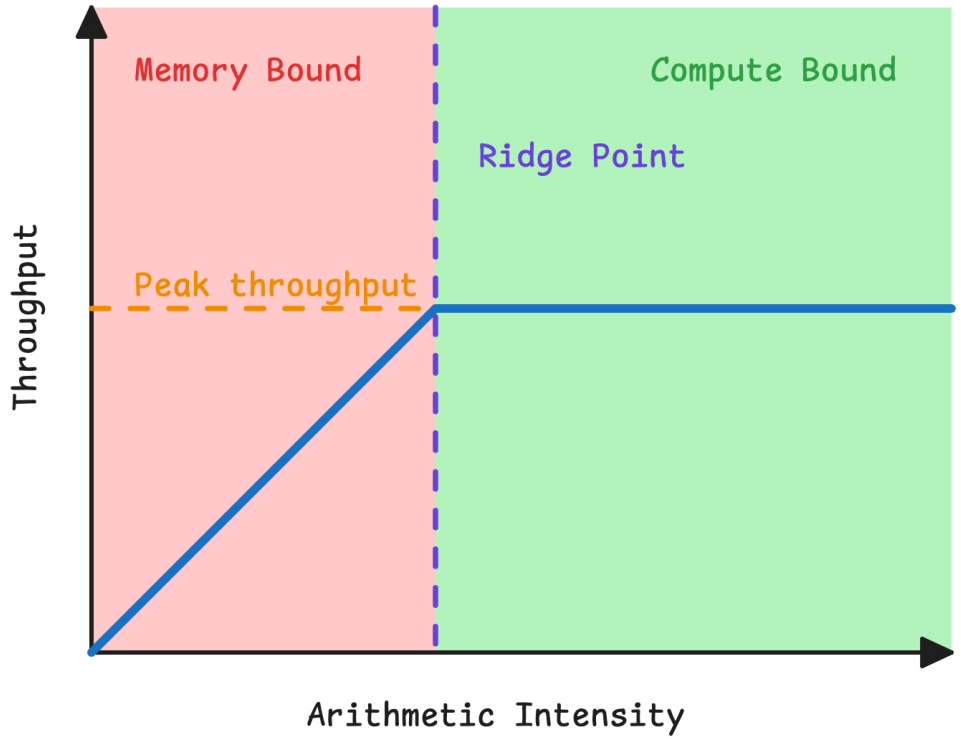
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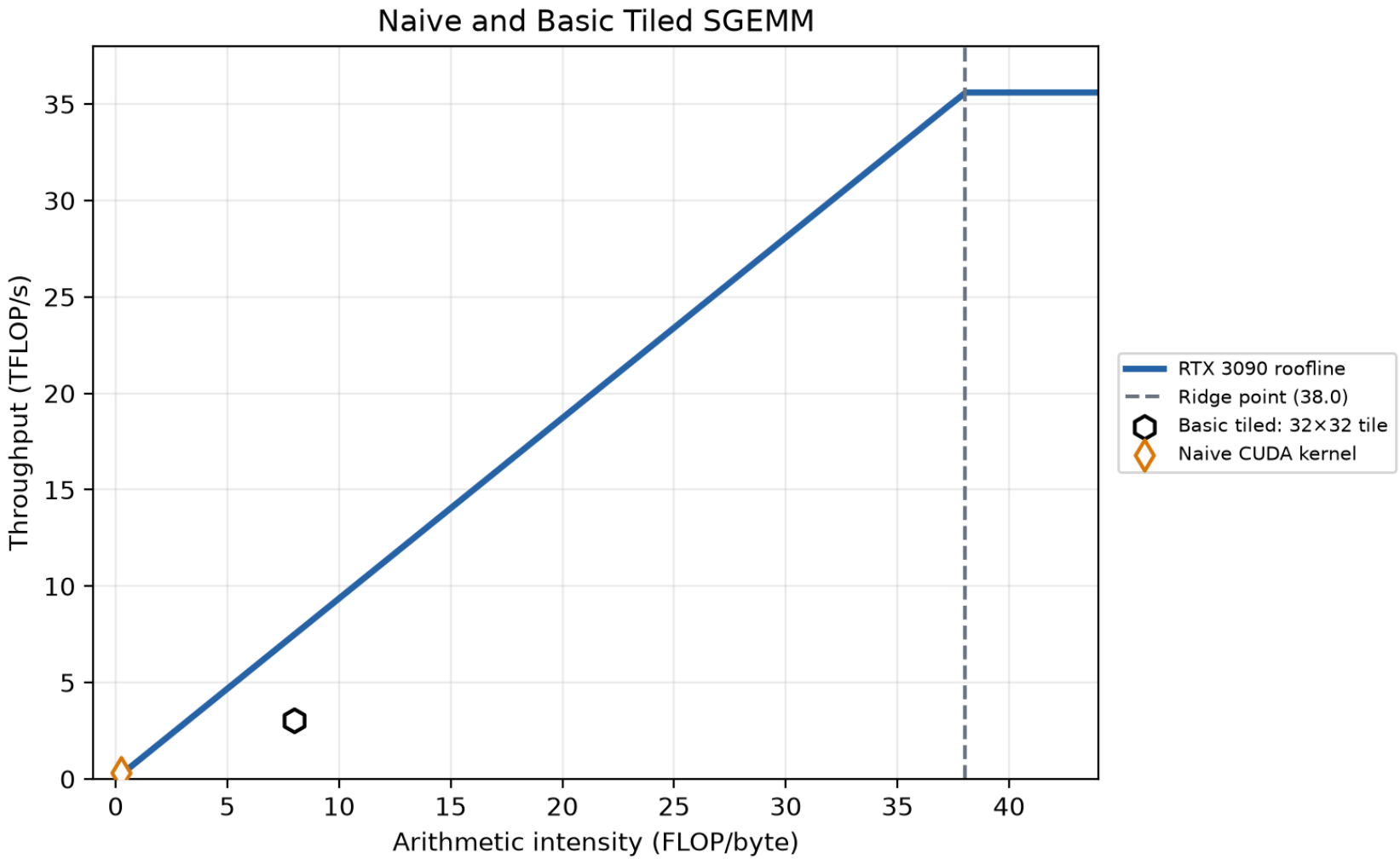
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Roofline Model

For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI. Up to the **ridge point**, we are **memory bound**, increasing AI will raise the upper bound of throughput. Past that, we are **compute bound**, increasing AI will not raise the upper bound of throughput.



Roofline Model of our RTX 3090

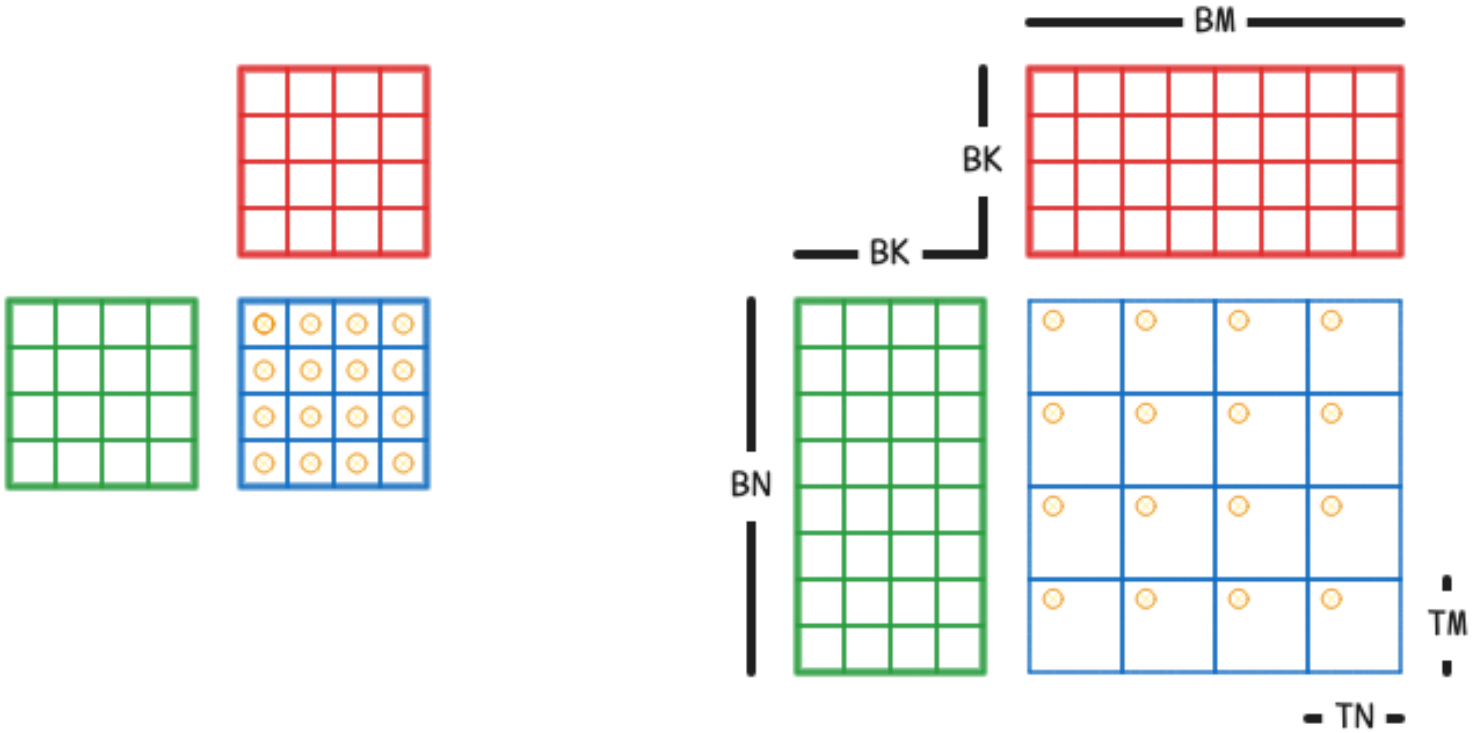


Section Overview

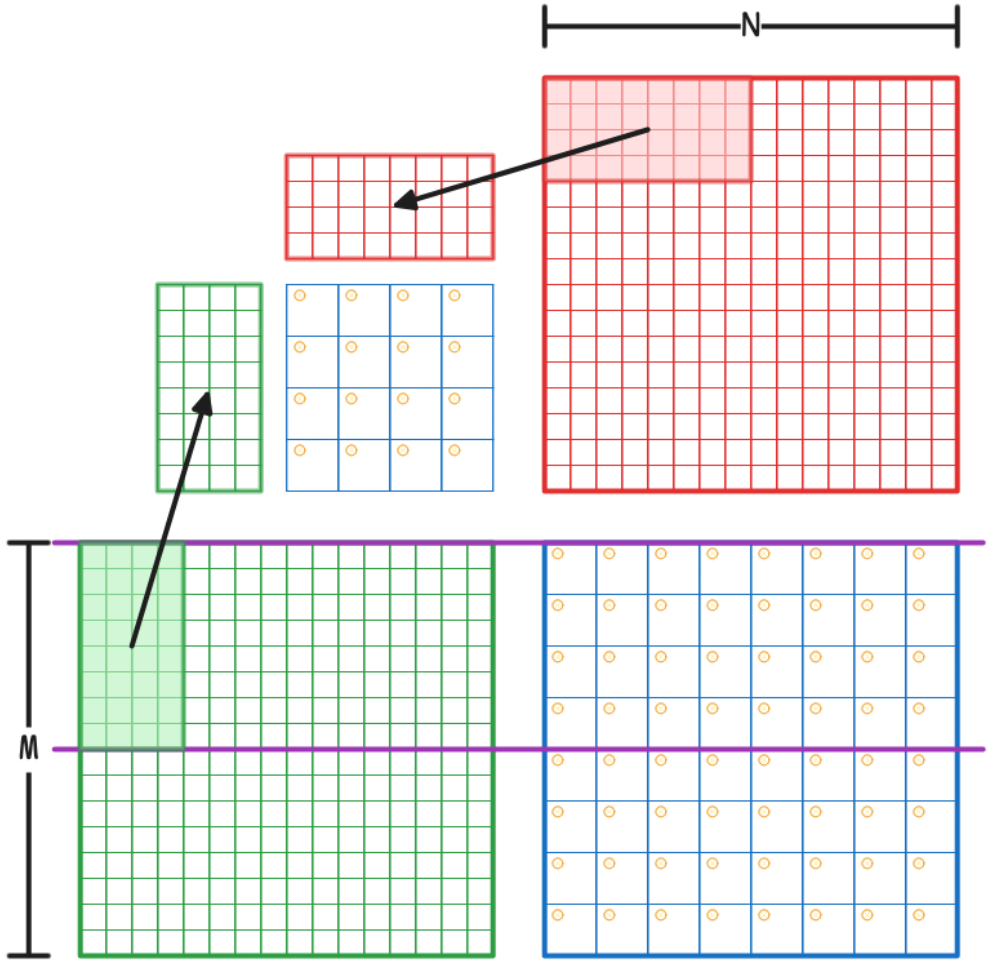
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Thread coarsening: Increasing AI

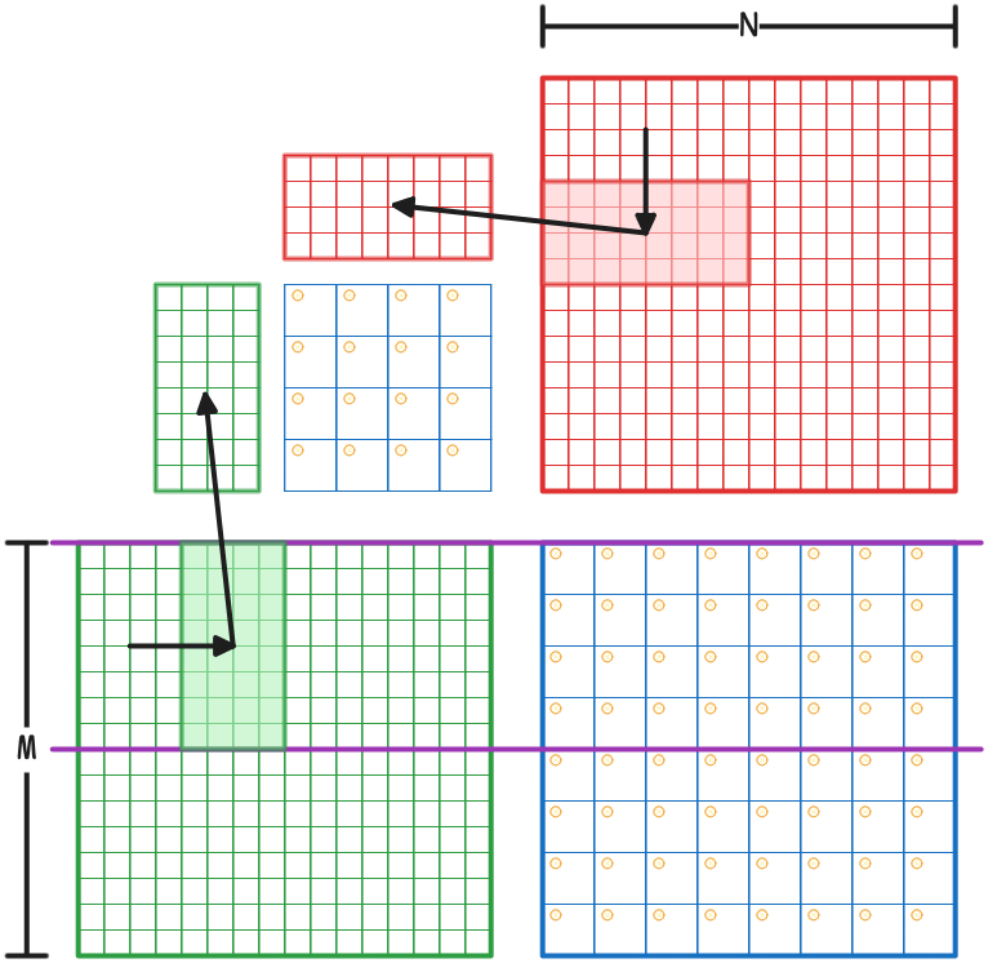
- Basic tiling assigns 1 element of the output to each thread.
- We can *coarsen* each thread, and assign a $TM \times TN$ portion of the output to each thread.
- We load larger chunks into SMEM, increasing data reuse and AI.
- One loaded A value can update several columns; one loaded B value can update several rows.



Coarsened SGEMM visualization



Coarsened SGEMM visualization



Coarsening factors are a tradeoff

Two levels of tiling

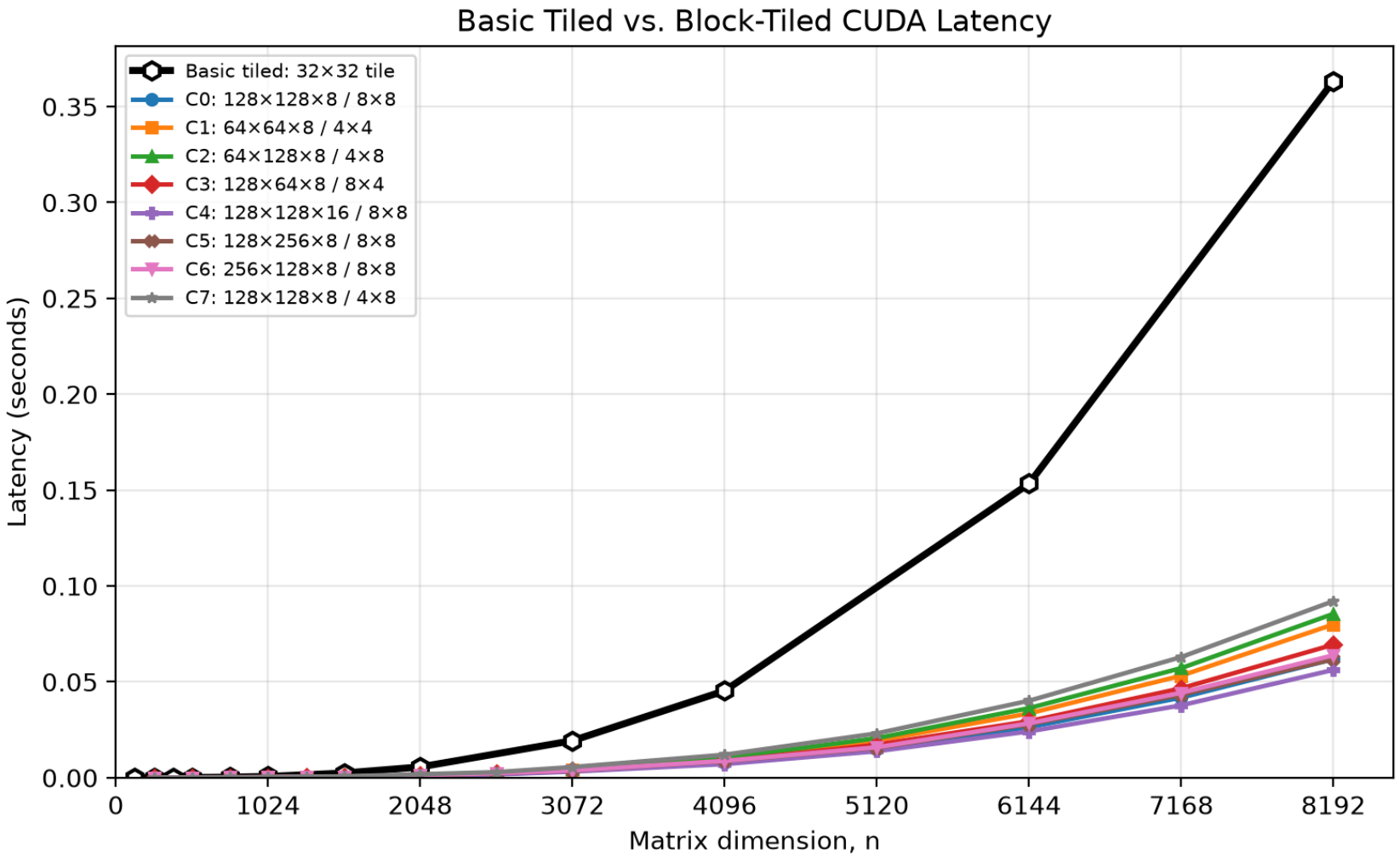
- A block computes a $BM \times BN$ output tile while advancing through K in steps of BK.
- Each thread computes a $TM \times TN$ micro-tile in registers.
- Thread count: $\frac{BM}{TM}\frac{BN}{TN}$.

Why not keep increasing them?

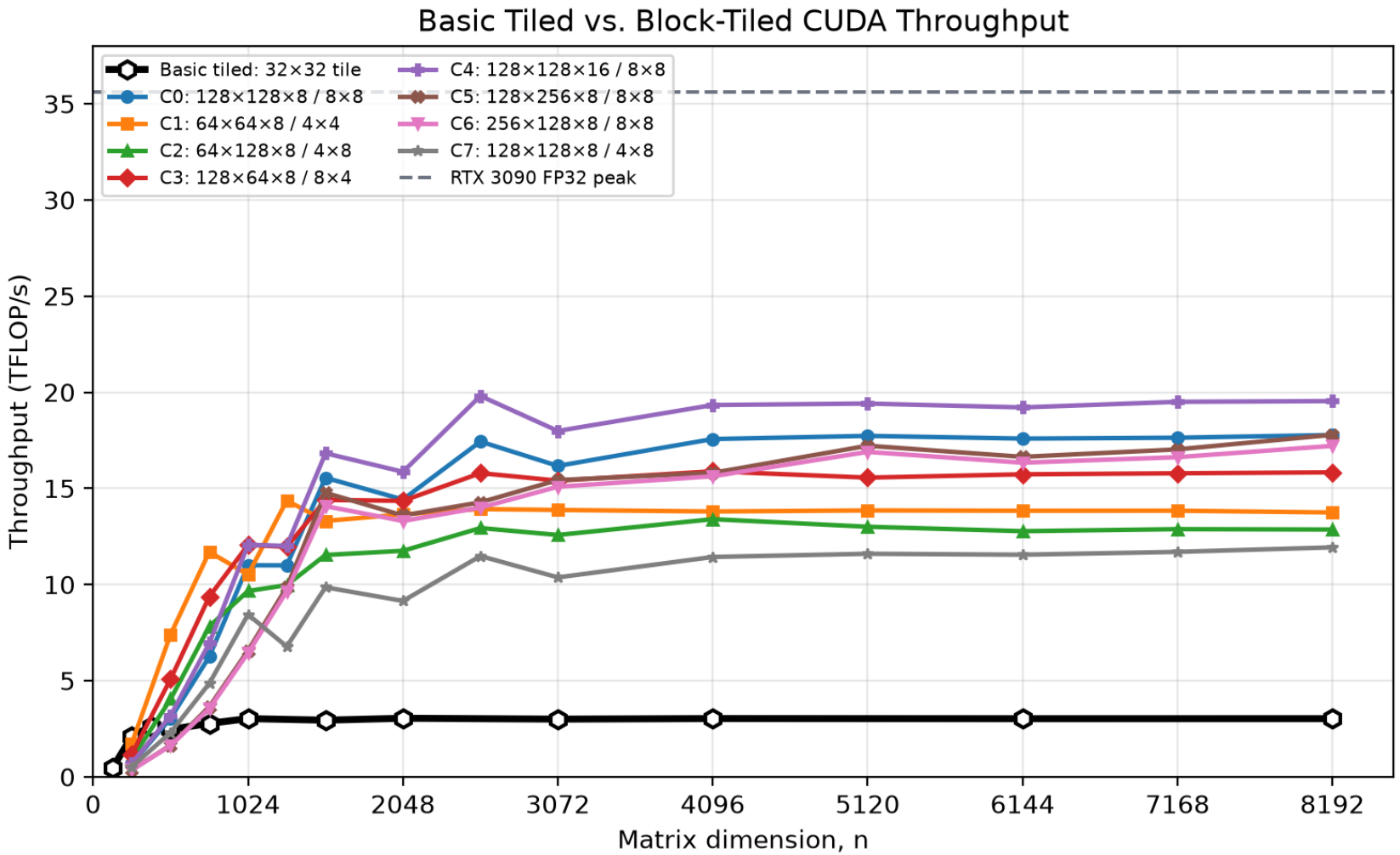
Larger tiles improve reuse and arithmetic intensity, but consume more registers and shared memory. Eventually occupancy falls and extra parallelism or reuse no longer helps.

Config.	$BM \times BN \times BK$	$TM \times TN$	AI	Throughput
C1	$64 \times 64 \times 8$	4×4	15.9	13.75 TFLOP/s
C4	$128 \times 128 \times 16$	8×8	31.5	**19.54 TFLOP/s**
C5	$128 \times 256 \times 8$	8×8	41.8	17.79 TFLOP/s

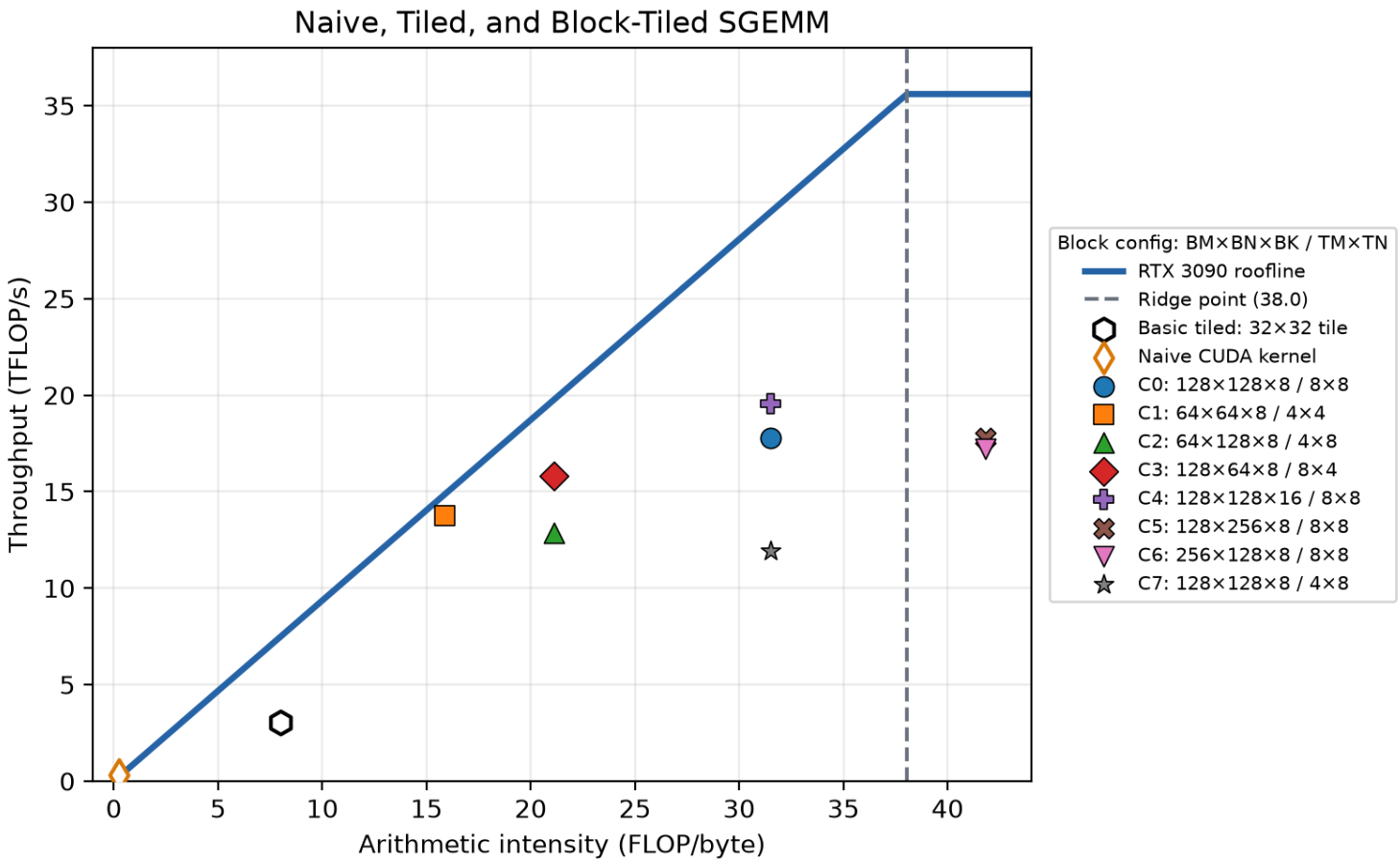
Basic tiled vs. block-tiled latency



Block tiling raises useful throughput



Tiling configurations on the roofline



Reminder that the Roofline Model only tells us the upper bound!

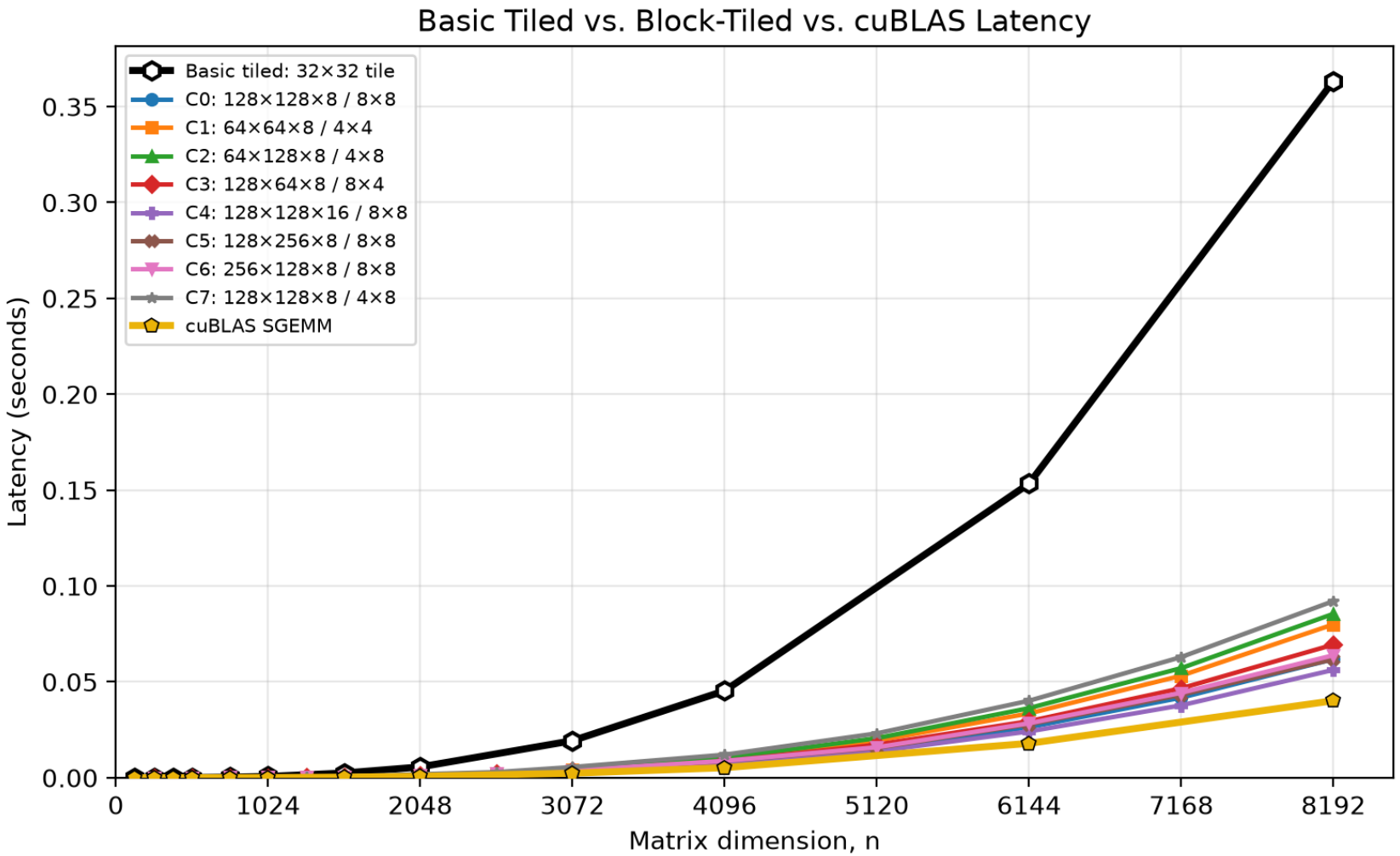
We have only scratched the surface: cuBLAS

- cuBLAS is NVIDIA's architecture-tuned linear algebra library and the practical performance target for our SGEMM kernels.
- At $n = 8192$, cuBLAS reaches 27.38 TFLOP/s, while our fastest configuration (C4) reaches 19.54 TFLOP/s, or 71.4% of cuBLAS's throughput.
- cuBLAS completes the multiplication in 40.2 ms versus 56.3 ms for C4, about 1.40× faster.

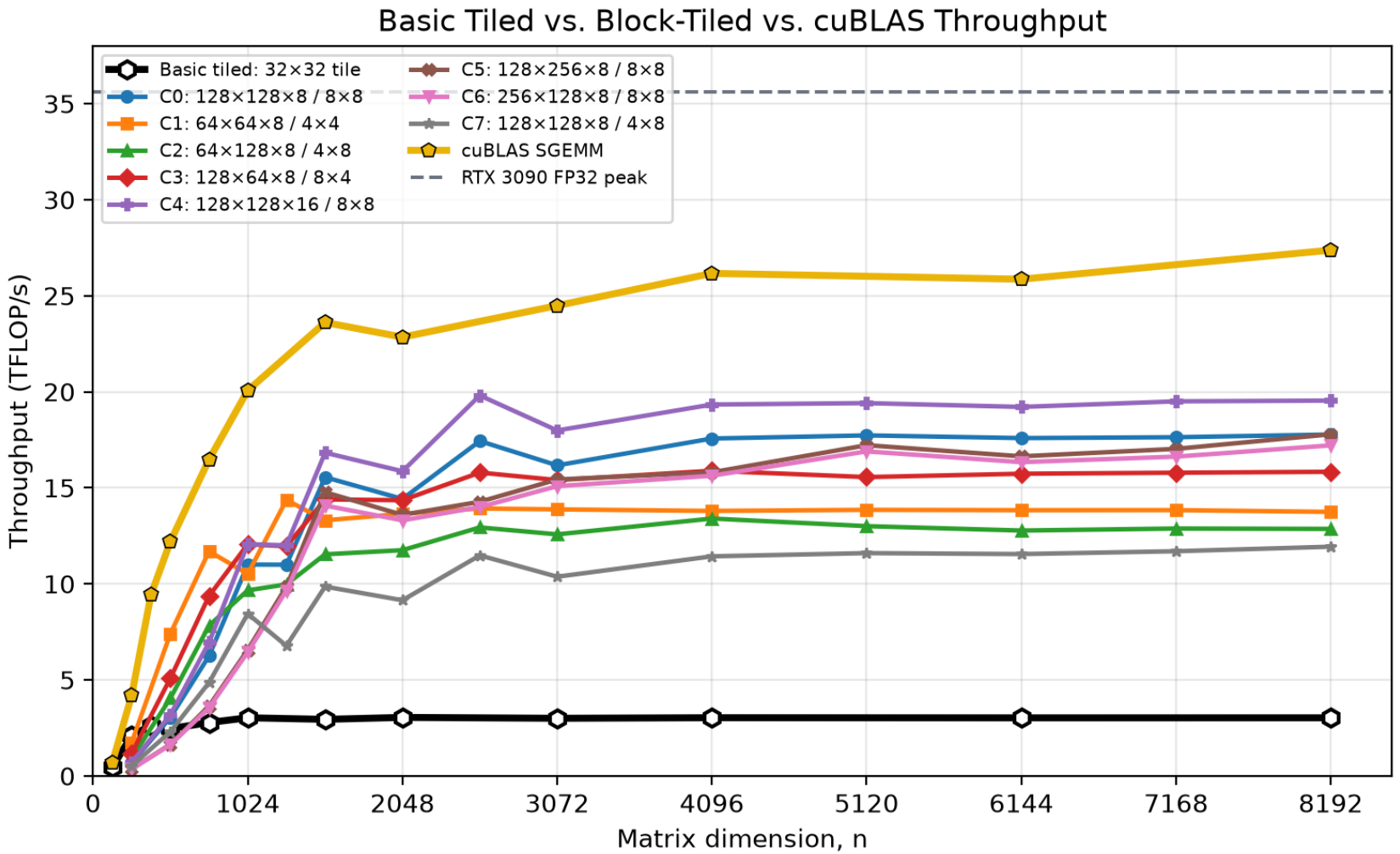
The remaining gap

Production kernels combine deeper register and warp tiling, vectorized memory access, pipelined loads, and architecture-specific instruction scheduling.

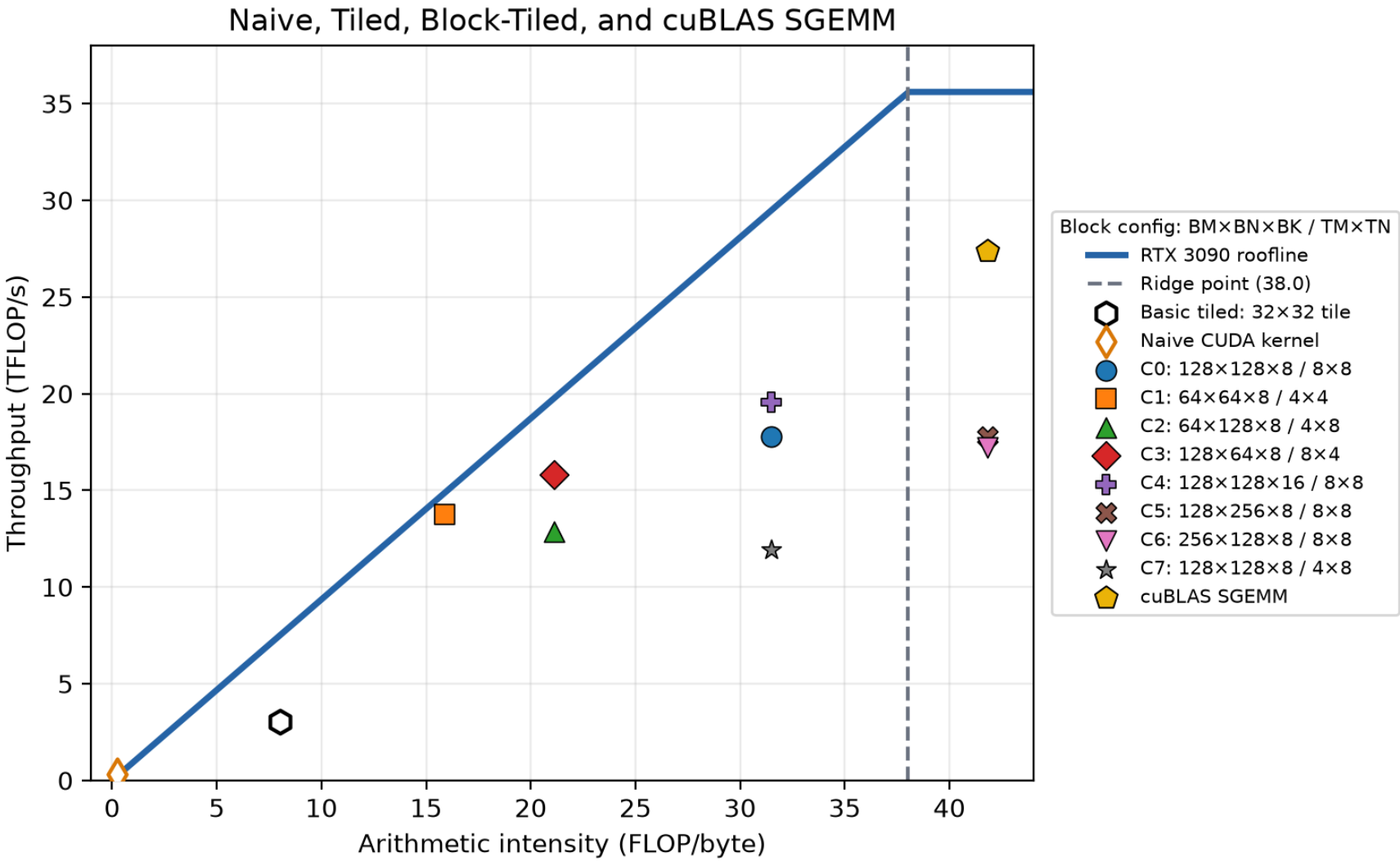
Basic tiled vs. block-tiled latency vs. cuBLAS



cuBLAS remains the performance target



cuBLAS on the roofline



cuBLAS arithmetic intensity is estimated from the profiled 256×128 output-tile shape.

We've come a long way

